MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE CHH

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic NAND gate module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table II.

| | | | |
|---|---|---|---|
| Name | Gate, NAND | | |
| Family | Digital | | |
| Key code | CHH | | |
| Size | 1A | | |
| Weight | 35 grams maximum | | |
| Environment | Class I or class II as specified in MIL-STD-1389. | | |
| | Class I | Class II | Unit |
| Failure rate | 0.085 | 0.173 | Failures /$10^8$ hours maximum |
| Maximum power dissipation; outputs not loaded ($V_{CC}$ = 5.5 V) | | | |
| All outputs at low level | 48 | 48 | mW |
| All outputs at high level | 18 | 18 | mW |
| Supply voltage | 5.0 Vdc ±10 percent | 5.0 Vdc ±10 percent | |
| Input voltage | | | |
| Low level | -0.5 min, 0.7 max | -0.5 min, 0.7 max | V |
| High level | 2.0 min, $V_{CC}$ max | 2.0 min, $V_{CC}$ max | V |
| Input current | | | |
| Low level ($V_{IL}$ = 0 V) | -0.434 max | -0.434 max | mA |
| ($V_{IL}$ = 0.4 V) | -0.4 max | -0.4 max | mA |
| High level ($V_{IH}$ = 2.5 V) | 20 max | 20 max | µA |
| ($V_{IH}$ = 7.0 V) | 100 max | 100 max | µA |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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| | Class I | Class II | Unit |
|---|---|---|---|
| Output voltage | | | |
| Low level ($I_{OL}$ = 4.0 mA) | 0.4 max | 0.4 max | V |
| High level ($I_{OH}$ = -400 µA) | 2.5 min | 2.5 min | V |
| Output drive current | | | |
| Low level ($V_{OL}$ = 0.4 V) | 4.0 max | 4.0 max | mA |
| High level ($V_{OH}$ = 2.0 V) | -400 min | -400 min | µA |
| Propagation delay time ($t_{PHL}$) | | | |
| $C_L$ = 50 pF | 3.0 min, 21.0 max | 3.0 min, 23.0 max | ns |
| $C_L$ = 150 pF | 4.0 min, 31.0 max | 4.0 min, 33.0 max | ns |
| Propagation delay time ($T_{PLH}$) | | | |
| $C_L$ = 50 pF | 3.0 min, 21.0 max | 3.0 min, 23.0 max | ns |
| $C_L$ = 150 pF | 4.0 min, 31.0 max | 4.0 min, 33.0 max | ns |
| Delta propagation delay time ($\Delta t_{PHL}$) ($C_L$ = 50 pF) | | | |
| Between gates 1 and 2 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Between gates 3 and 4 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Between gates 5 and 6 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Between gates 7 and 8 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Delta propagation delay time ($\Delta t_{PLH}$) ($C_L$ = 50 pF) | | | |
| Between gates 1 and 2 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Between gates 3 and 4 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Between gates 5 and 6 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |
| Between gates 7 and 8 | -3.0 min, 3.0 max | -3.0 min, 3.0 max | ns |

1.3 Module function. This module contains four independent 3-input NAND gates and four independent 4-input NAND gates.

1.3.1 Input to output relationship. Each NAND gate has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

| 3-input NAND gate | | | | 4-input NAND gate | | | | |
|---|---|---|---|---|---|---|---|---|
| Inputs | | | Output | Inputs | | | | Output |
| A | B | C | D | A | B | C | D | E |
| L | L | L | H | L | L | L | L | H |
| L | L | H | H | L | L | L | H | H |
| L | H | L | H | L | L | H | L | H |
| L | H | H | H | L | L | H | H | H |
| H | L | L | H | L | H | L | L | H |
| H | L | H | H | L | H | L | H | H |
| H | H | L | H | L | H | H | L | H |
| H | H | H | L | L | H | H | H | H |
| | | | | H | L | L | L | H |
| | | | | H | L | L | H | H |
| | | | | H | L | H | L | H |
| | | | | H | L | H | H | H |
| | | | | H | H | L | L | H |
| | | | | H | H | L | H | H |
| | | | | H | H | H | L | H |
| | | | | H | H | H | H | L |

NOTES:
1. H = High-level voltage.
2. L = Low-level voltage.



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1.3.2 Ground pins. All electrical parameters are measured with pins 3, 10, and 18 connected separately to the 0 volt return. System problems may result from a different grounding method.

1.3.3 Parallel outputs. Only the following outputs may be paralleled provided their corresponding inputs are parallel:

a. Outputs from gates 1 and 2 may be paralleled.
b. Outputs from gates 3 and 4 may be paralleled.
c. Outputs from gates 5 and 6 may be paralleled.
d. Outputs from gates 7 and 8 may be paralleled.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.085 failures per million operating hours for class I or 0.173 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CHH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.



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3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/293-1 for class I and M28787/293-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

| Equipment | MIL-STD-1665 designator | No. required (maximum per test figure) |
|---|---|---|
| DC power supply | Item P01 | 4 |
| Pulse generator | Item J01 | 1 |
| Megohm bridge | Item R06 | 1 |
| DC voltmeter | Item V03 | 1 |
| DC ammeter | Item X01 | 1 |
| Resistance bridge | Item R02 | 1 |
| Oscilloscope | Item D05 | 1 |
| Oscilloscope probe | Item V01 | 2 |
| Pulse generator | Item J05 | 1 |
| Capacitance meter | Item C05 | 1 |
| Capacitance bridge | Item C01 | 1 |
| **Resistors** | | |
| 1.27 kΩ ±1 percent, 1/4 W | | 8 |
| 1.0 kΩ ±1 percent, 1/4 W | | 1 |
| 51Ω ±5 percent, 1/2 W | | 2 |
| 750Ω ±1 percent, 1/8 W | | 1 |
| Value as required for figure 12, note 2, ±1 percent, 1/8 W | | 8 |
| 6.19 kΩ ±1 percent, 1/8 W | | 8 |



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| Equipment | MIL-STD-1665 designator | No. required (maximum per test figure) |
|---|---|---|
| Capacitors | | |
| Value as required for figure 12, note 4, test type S | | 8 |
| Value as required for figure 12, note 4, test type Q | | 1 |
| Diode | | |
| 1N4150, or equivalent | | |

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

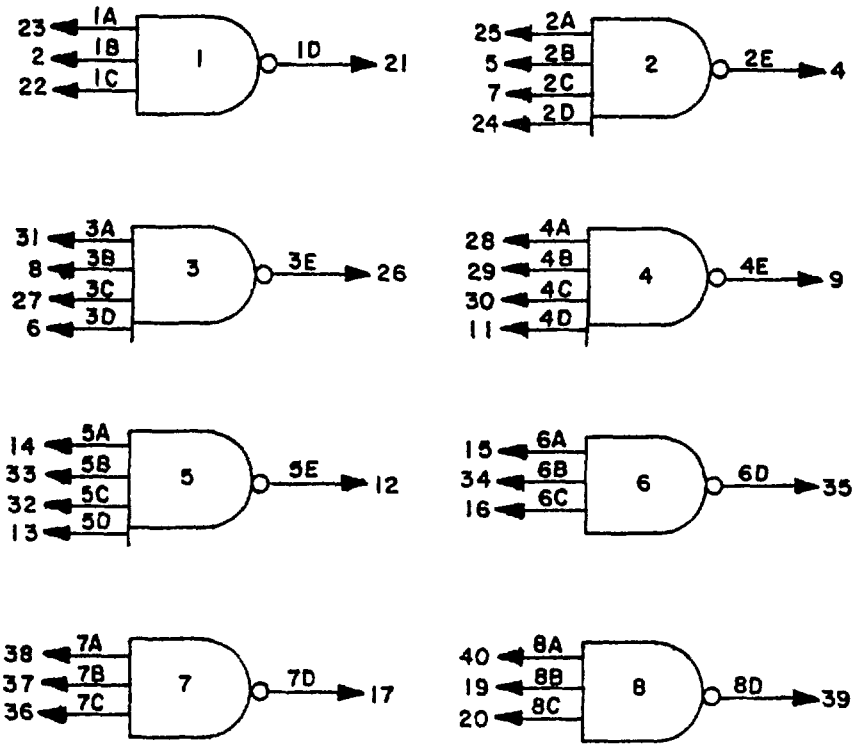
5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

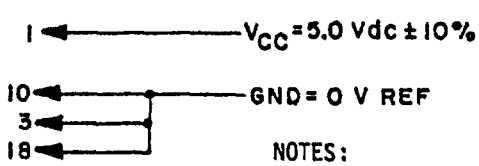
6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.



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NOTES:
1. 1A-8A = Gate inputs.
2. 1B-8B = Gate inputs.
3. 1C-8C = Gate inputs.
4. 2D-5D = Gate inputs.
5. 1D, 6D, 7D, 8D = Gate outputs.
6. 2E-5E = Gate outputs

FIGURE 1. CHH module function diagram, digital NAND gate.



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TABLE II. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25° ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0° ±5°C | | 60° ±5°C | | -55° ±5°C | | 100° ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| A | Isolation<br>Initial<br>End-of-life | III | S | <br>10<br>1.0 | <br>-<br>- | <br>10<br>1.0 | <br>-<br>- | <br>10<br>1.0 | <br>-<br>- | <br>10<br>1.0 | <br>-<br>- | <br>10<br>1.0 | <br>-<br>- | <br>MΩ<br>MΩ |
| B1-B8 | $V_{OUT}$ (low-loaded) | IV | S | 0.01 | 0.8 | 0.01 | 0.8 | 0.01 | 0.8 | 0.01 | 0.8 | 0.01 | 0.8 | V |
| C1-C28 | Input breakdown current | V | S | 0.0 | 100 | 0.0 | 100 | 0.0 | 100 | 0.0 | 100 | 0.0 | 100 | µA |
| D1-D8 | Output short circuit current | VI | X | -15 | -100 | -15 | -100 | -15 | -100 | -15 | -100 | -15 | -100 | mA |
| E1-E8 | Output leakage current | VII | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | mA |
| F1-F28 | Input clamp diode voltage | VIII | X | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | V |
| G1-G8 | Static output voltage, low level | IX | X | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | V |
| H1-H28 | Static output voltage, high level | X | X | 2.5 | 4.5 | 2.5 | 4.5 | 2.5 | 4.5 | 2.5 | 4.5 | 2.5 | 4.5 | V |
| J1-J28 | Input current, high level | XI | X | 0 | 20 | 0 | 20 | 0 | 20 | 0 | 20 | 0 | 20 | µA |
| K1-K28 | Input current, low level | XII | X | -0.01 | -0.434 | -0.01 | -0.434 | 0.01 | -0.434 | -0.01 | -0.434 | -0.01 | -0.434 | mA |
| L1, L2 | Continuity | XIII | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ω |
| L3, L4 | Continuity | XIII | Q | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | Ω |
| M1 | Power supply current (low-level) outputs | XIV | S | 0.1 | 8.8 | 0.1 | 8.8 | 0.1 | 8.8 | 0.1 | 8.8 | 0.1 | 8.8 | mA |
| M2 | Power supply current (high-level) outputs | XIV | S | 0.1 | 3.2 | 0.1 | 3.2 | 0.1 | 3.2 | 0.1 | 3.2 | 0.1 | 3.2 | mA |
| N1-N8 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | XV | S | 5.0 | 50.0 | 5.0 | 50.0 | 5.0 | 55.0 | 5.0 | 50.0 | 5.0 | 60.0 | ns |

See footnotes at end of table.





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TABLE II. Module electrical requirements -Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25° ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0° ±5°C | | 60° ±5°C | | -55° ±5°C | | 100° ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| N9-N16 | Transition time ($t_{TLH}$), $C_L$ = 150 pF | XV | Q | 10.0 | 80.0 | 10.0 | 80.0 | 10.0 | 90.0 | 10.0 | 80.0 | 10.0 | 100.0 | ns |
| P1-P8 | Transition time ($t_{THL}$), $C_L$ = 50 pF | XV | S | 3.0 | 18.0 | 3.0 | 18.0 | 3.0 | 20.0 | 3.0 | 18.0 | 3.0 | 22.0 | ns |
| P9-P16 | Transition time ($t_{THL}$), $C_L$ = 150 pF | XV | Q | 6.0 | 40.0 | 6.0 | 40.0 | 6.0 | 42.0 | 6.0 | 40.0 | 6.0 | 44.0 | ns |
| Q1-Q28 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XV | S | 3.0 | 19.0 | 3.0 | 19.0 | 3.0 | 21.0 | 3.0 | 19.0 | 3.0 | 23.0 | ns |
| Q29-Q56 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XV | Q | 4.0 | 29.0 | 4.0 | 29.0 | 4.0 | 31.0 | 4.0 | 29.0 | 4.0 | 33.0 | ns |
| R1-R28 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XV | S | 3.0 | 19.0 | 3.0 | 19.0 | 3.0 | 21.0 | 3.0 | 19.0 | 3.0 | 23.0 | ns |
| R29-R56 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XV | Q | 4.0 | 29.0 | 4.0 | 29.0 | 4.0 | 31.0 | 4.0 | 29.0 | 4.0 | 33.0 | ns |
| S1-S4 | Output to output delta propagation delay time ($\Delta t_{PLH}$), $C_L$ = 50 pF | XVI | S | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | ns |
| T1-T4 | Output to output delta propagation delay time ($\Delta t_{PHL}$), $C_L$ = 50 pF | XVI | S | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | ns |
| U1-U8 | Output crosstalk, low level | XVII | Q | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| V1-V8 | Output crosstalk, high level | XVII | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |

See footnotes at end of table.





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TABLE II. Module electrical requirements -Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25° ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0° ±5°C | | 60° ±5°C | | -55° ±5°C | | 100° ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| W | Power supply transient amplitude | XVIII | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| X | Power supply filter capacitance | XIX | S | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | µF |
| Y1-Y28 | Input capacitance | XX | Q | 5.0 | 15.0 | 5.0 | 15.0 | 5.0 | 15.0 | 5.0 | 15.0 | 5.0 | 15.0 | pF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.





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6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

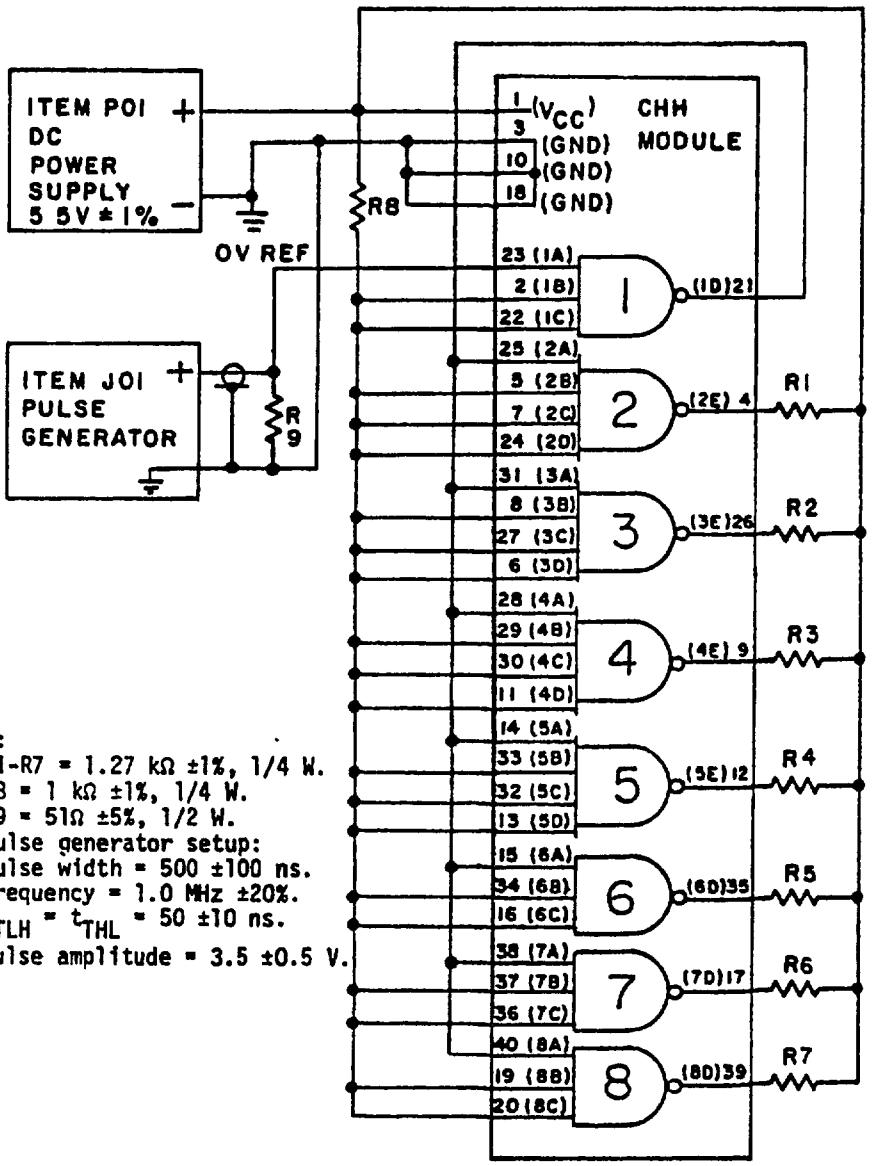
Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N082)



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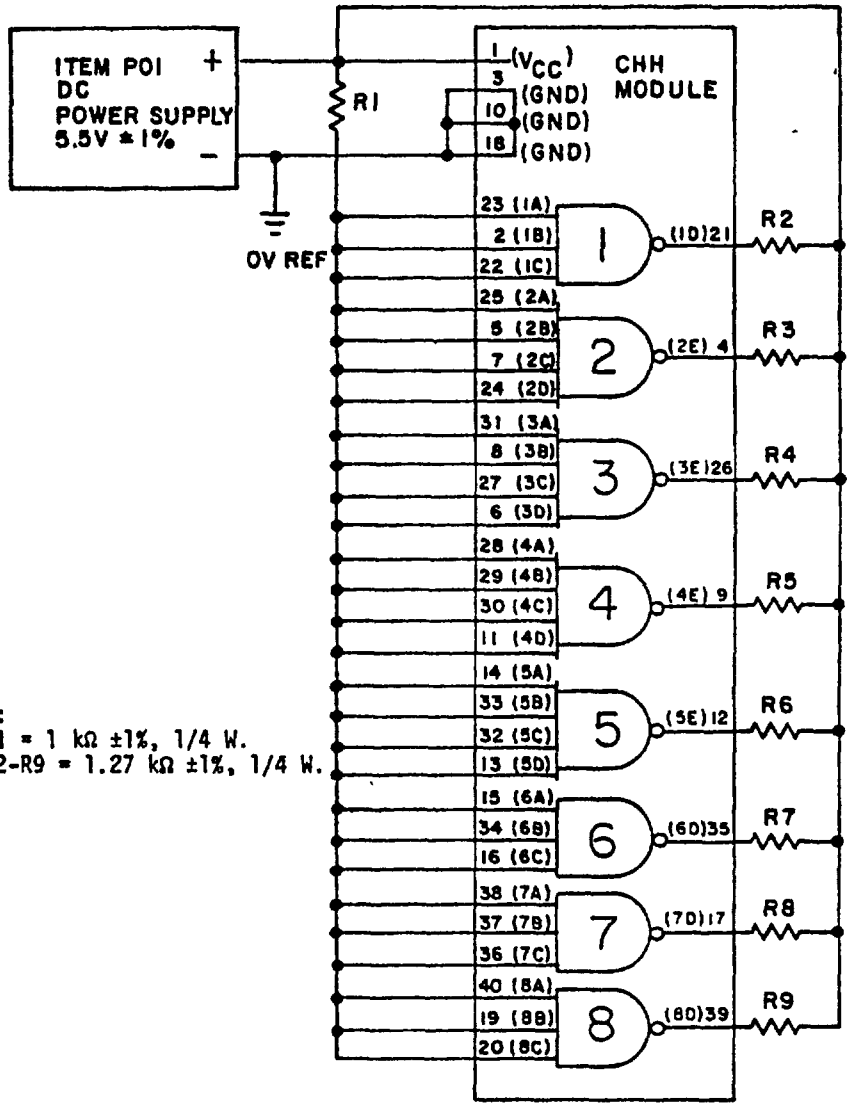



NOTES:
1. R1-R7 = 1.27 kΩ ±1%, 1/4 W.
2. R8 = 1 kΩ ±1%, 1/4 W.
3. R9 = 51Ω ±5%, 1/2 W.
4. Pulse generator setup:
   Pulse width = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL} = 50 \pm 10$ ns.
   Pulse amplitude = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.



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NOTES:
1. R1 = 1 kΩ ±1%, 1/4 W.
2. R2-R9 = 1.27 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.



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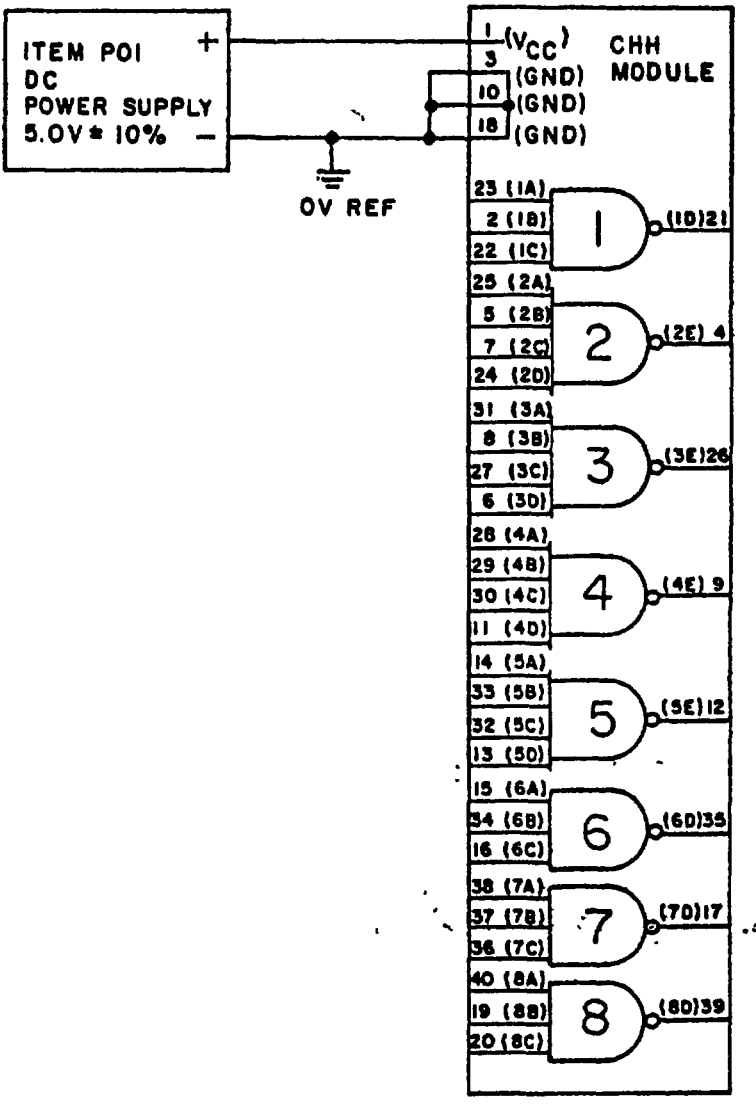



FIGURE 4. Vibration test circuit.



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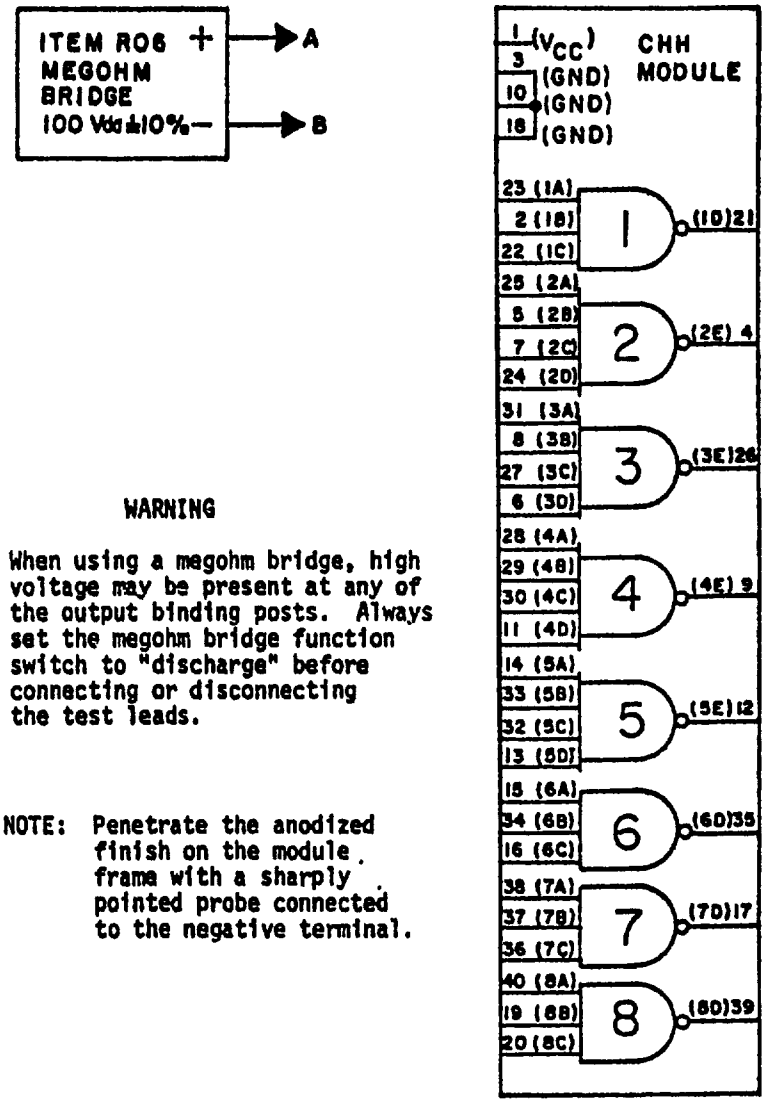
TABLE III. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A | Isolation | All pins | Frame | 10 | ∞ | MΩ | 5 |





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**WARNING**

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.



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TABLE IV. $V_{OUT}$ (low-loaded) test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| B1 | 1D $V_{OUT}$ (low loaded) | 21 | 0.01 | 0.8 | V | 6 |
| B2 | 2E | 4 | | | | |
| B3 | 3E | 26 | | | | |
| B4 | 4E | 9 | | | | |
| B5 | 5E | 12 | | | | |
| B6 | 6D | 35 | | | | |
| B7 | 7D | 17 | | | | |
| B8 | 8D $V_{OUT}$ (low loaded) | 39 | 0.01 | 0.8 | V | 6 |





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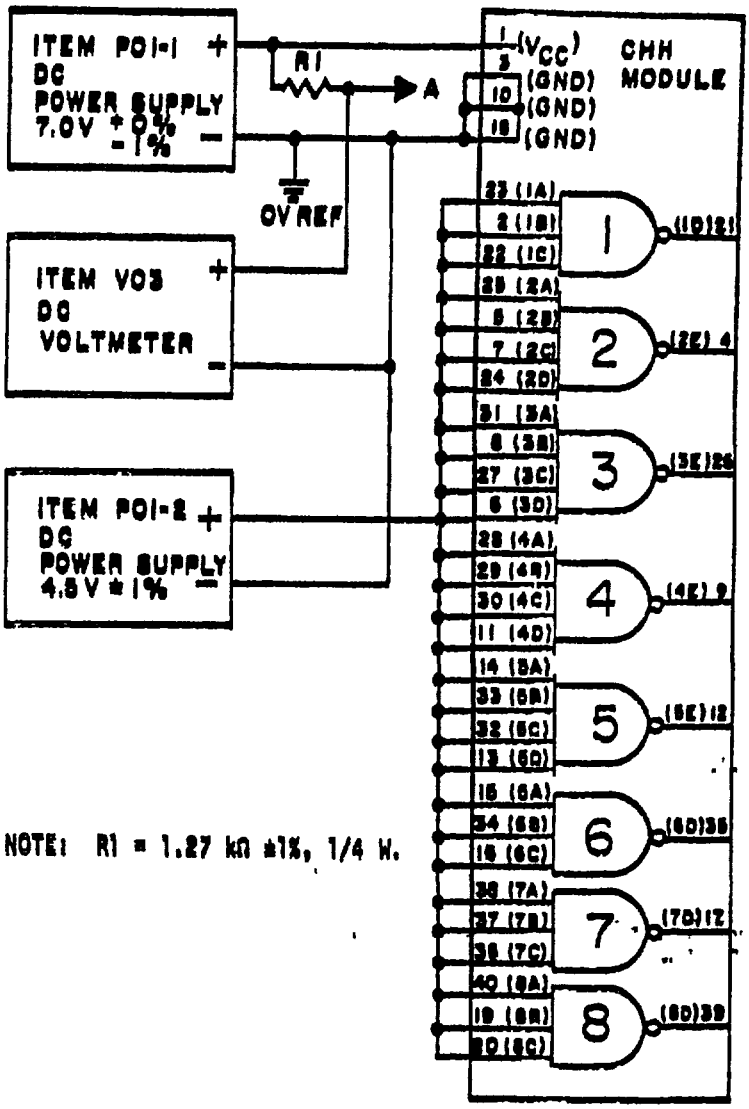



FIGURE 6. $V_{OUT}$ (low-loaded) test circuit for table IV.



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TABLE V. Input breakdown current test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C1 | 1A Input breakdown current | 23 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |
| C2 | 1B | 2 | 5,6,7,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C3 | 1C | 22 | 2,5,6,7,8,11,13,14,15,16, 19,20,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C4 | 2A | 25 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C5 | 2B | 5 | 2,6,7,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C6 | 2C | 7 | 2,5,6,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C7 | 2D | 24 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C8 | 3A | 31 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,32,33,34,36,37,38,40 | | | | |
| C9 | 3B Input breakdown current | 8 | 2,5,6,7,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |





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TABLE V. Input breakdown current test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C10 | 3C Input breakdown current | 27 | 2,5,6,7,8,11,13,14,15,16,19, 20,22,23,24,25,28,29,30,31, 32,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |
| C11 | 3D | 6 | 2,5,7,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C12 | 4A | 28 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C13 | 4B | 29 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,30, 31,32,33,34,36,37,38,40 | | | | |
| C14 | 4C | 30 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 31,32,33,34,36,37,38,40 | | | | |
| C15 | 4D | 11 | 2,5,6,7,8,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C16 | 5A | 14 | 2,5,6,7,8,11,13,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C17 | 5B | 33 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,34,36,37,38,40 | | | | |
| C18 | 5C Input breakdown current | 32 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |





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TABLE V. Input breakdown current test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C19 | 5D Input breakdown current | 13 | 2,5,6,7,8,11,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |
| C20 | 6A | 15 | 2,5,6,7,8,11,13,14,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C21 | 6B | 34 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,36,37,38,40 | | | | |
| C22 | 6C | 16 | 2,5,6,7,8,11,13,14,15,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| C23 | 7A | 38 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,40 | | | | |
| C24 | 7B | 37 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,38,40 | | | | |
| C25 | 7C | 36 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,37,38,40 | | | | |
| C26 | 8A | 40 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38 | | | | |
| C27 | 8B Input breakdown current | 19 | 2,5,6,7,8,11,13,14,15,16, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |





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TABLE V. Input breakdown current test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C28 | 8C Input breakdown current | 20 | 2,5,6,7,8,11,13,14,15,16, 19,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0.0 | 100 | µA | 7 |



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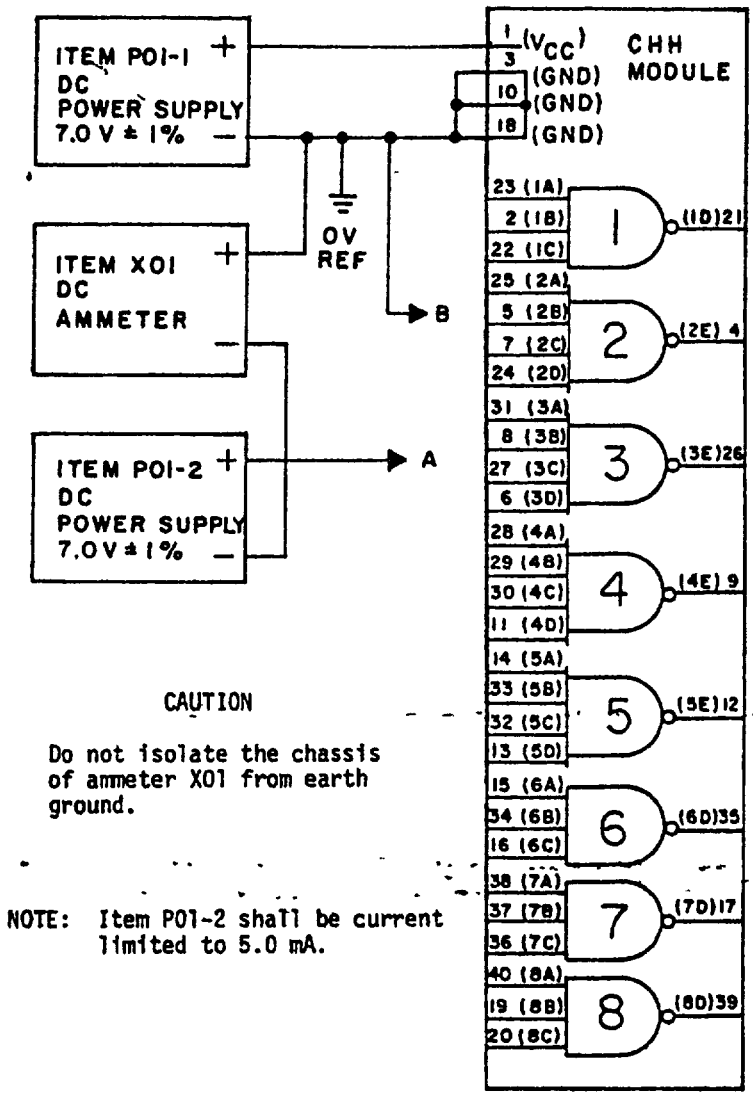



FIGURE 7. Input breakdown current test circuit for table V.



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TABLE VI. Output short circuit current test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| D1 | 1D Output short circuit current | 21 | -15 | -100 | mA | 8 |
| D2 | 2E | 4 | | | | |
| D3 | 3E | 26 | | | | |
| D4 | 4E | 9 | | | | |
| D5 | 5E | 12 | | | | |
| D6 | 6D | 35 | | | | |
| D7 | 7D | 17 | | | | |
| D8 | 8D Output short circuit current | 39 | -15 | -100 | mA | 8 |





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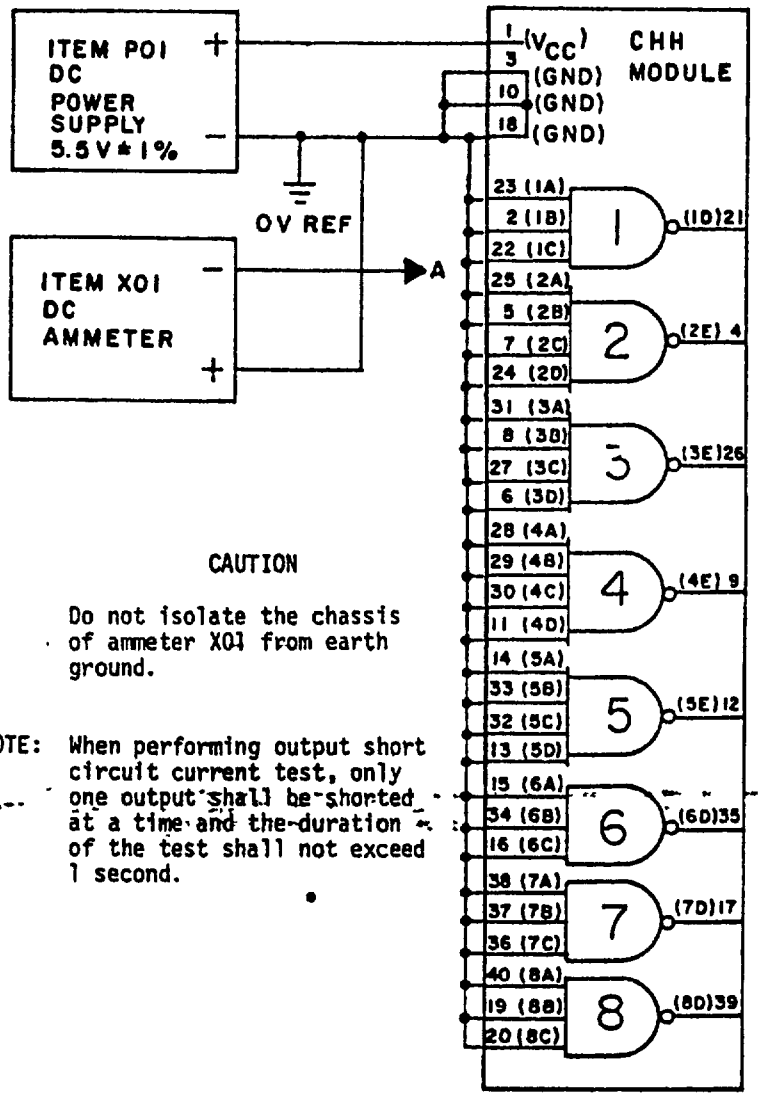



FIGURE 8. Output short circuit current test circuit for table VI.



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TABLE VII. Output leakage current test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| E1 | 1D Output leakage current | 21 | 0.0 | 1.0 | mA | 9 |
| E2 | 2E | 4 | | | | |
| E3 | 3E | 26 | | | | |
| E4 | 4E | 9 | | | | |
| E5 | 5E | 12 | | | | |
| E6 | 6D | 35 | | | | |
| E7 | 7D | 17 | | | | |
| E8 | 8D Output leakage current | 39 | 0.0 | 1.0 | mA | 9 |





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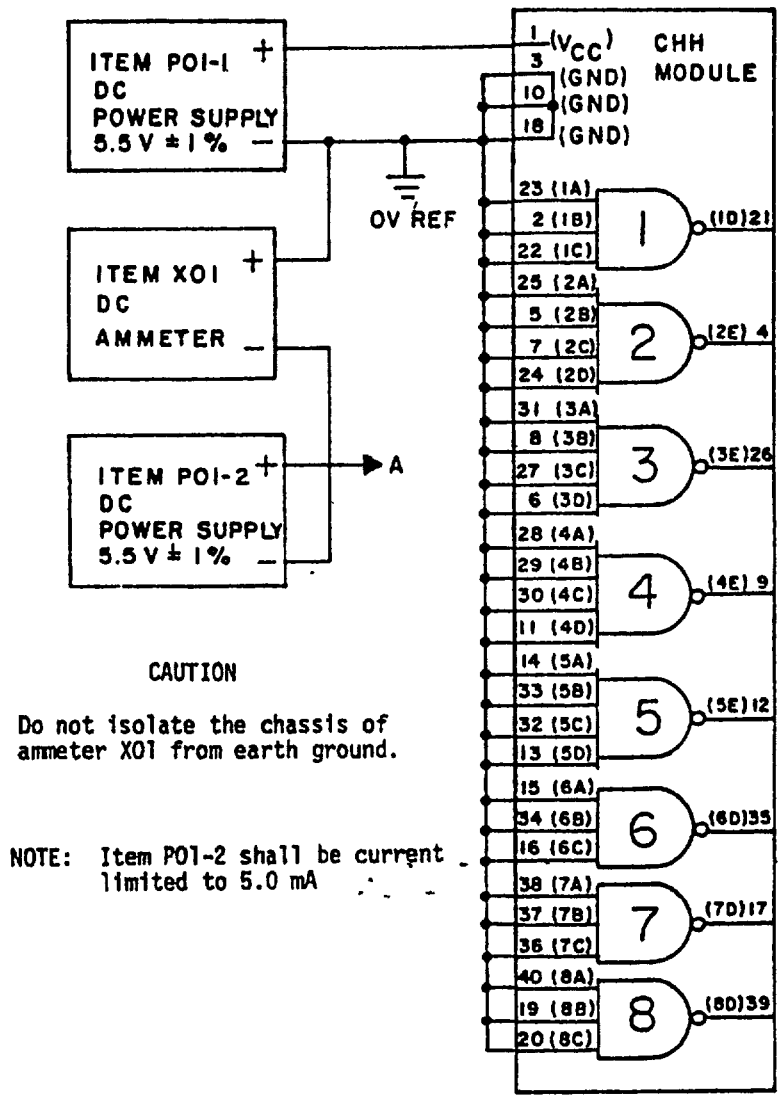



FIGURE 9. Output leakage current test circuit for table VII.



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TABLE VIII. Input clamp diode voltage test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| F1 | 1A Input clamp diode voltage | 23 | -0.4 | -1.5 | V | 10 |
| F2 | 1B | 2 | | | | |
| F3 | 1C | 22 | | | | |
| F4 | 2A | 25 | | | | |
| F5 | 2B | 5 | | | | |
| F6 | 2C | 7 | | | | |
| F7 | 2D | 24 | | | | |
| F8 | 3A | 31 | | | | |
| F9 | 3B | 8 | | | | |
| F10 | 3C | 27 | | | | |
| F11 | 3D | 6 | | | | |
| F12 | 4A | 28 | | | | |
| F13 | 4B | 29 | | | | |
| F14 | 4C | 30 | | | | |
| F15 | 4D | 11 | | | | |
| F16 | 5A | 14 | | | | |
| F17 | 5B | 33 | | | | |
| F18 | 5C | 32 | | | | |
| F19 | 5D Input clamp diode voltage | 13 | -0.4 | -1.5 | V | 10 |





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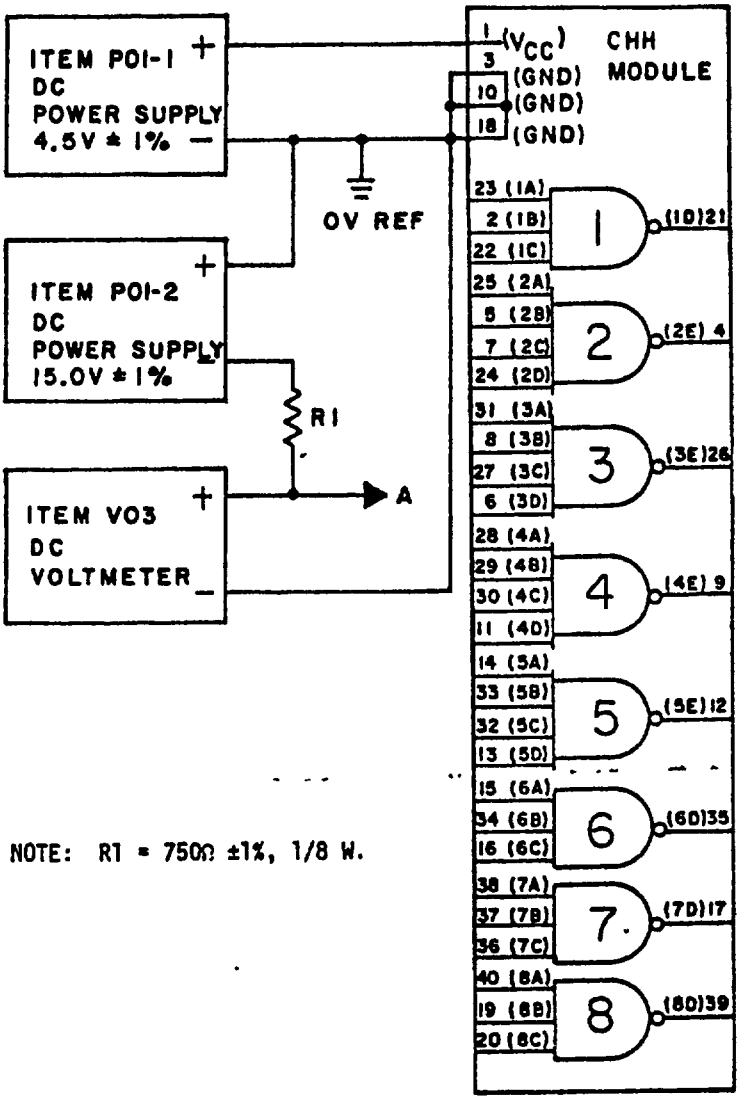
TABLE VIII. Input clamp diode voltage test - Continued.

| Test condition | Parameter | Connect O, A to pin | Initial limits 25° ±5°C Ambient Min | Initial limits 25° ±5°C Ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|
| F20 | 6A Input clamp diode voltage | 15 | -0.4 | -1.5 | V | 10 |
| F21 | 6B | 34 | | | | |
| F22 | 6C | 16 | | | | |
| F23 | 7A | 38 | | | | |
| F24 | 7B | 37 | | | | |
| F25 | 7C | 36 | | | | |
| F26 | 8A | 40 | | | | |
| F27 | 8B | 19 | | | | |
| F28 | 8C Input clamp diode voltage | 20 | -0.4 | -1.5 | V | 10 |





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NOTE: R1 = 750Ω ±1%, 1/8 W.

FIGURE 10. Input clamp diode voltage test circuit for table VIII.



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TABLE IX. Static output voltage (low-level) test.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| G1 | 1D Static output voltage, low level | 2,22,23 | Open | 21 | 0.0 | 0.4 | V | 11 |
| G2 | 2E | 5,7,24,25 | | 4 | | | | |
| G3 | 3E | 6,8,27,31 | | 26 | | | | |
| G4 | 4E | 28,29,30, 11 | | 9 | | | | |
| G5 | 5E | 13,14,32, 33 | | 12 | | | | |
| G6 | 6D | 15,16,34 | | 35 | | | | |
| G7 | 7D | 36,37,38 | | 17 | | | | |
| G8 | 8D Static output voltage, low level | 19,20,40 | Open | 39 | 0.0 | 0.4 | V | 11 |





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TABLE X. Static output voltage (high-level) test.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| H1 | 1D Static output voltage, high level | 22,23 | 2 | 21 | 2.5 | 4.5 | V | 11 |
| H2 | 1D | 22,2 | 23 | 21 | | | | |
| H3 | 1D | 2,23 | 22 | 21 | | | | |
| H4 | 2E | 7,24,25 | 5 | 4 | | | | |
| H5 | 2E | 5,24,25 | 7 | | | | | |
| H6 | 2E | 5,7,25 | 24 | | | | | |
| H7 | 2E | 5,7,24 | 25 | 4 | | | | |
| H8 | 3E | 8,27,31 | 6 | 26 | | | | |
| H9 | 3E | 6,27,31 | 8 | | | | | |
| H10 | 3E | 6,8,31 | 27 | | | | | |
| H11 | 3E | 6,8,27 | 31 | 26 | | | | |
| H12 | 4E | 28,29,30 | 11 | 9 | | | | |
| H13 | 4E | 11,29,30 | 28 | | | | | |
| H14 | 4E | 11,28,30 | 29 | | | | | |
| H15 | 4E | 11,28,29 | 30 | 9 | | | | |
| H16 | 5E | 13,14,33 | 32 | 12 | | | | |
| H17 | 5E | 13,14,32 | 33 | 12 | | | | |
| H18 | 5E Static output voltage, high level | 14,33,32 | 13 | 12 | 2.5 | 4.5 | V | 11 |





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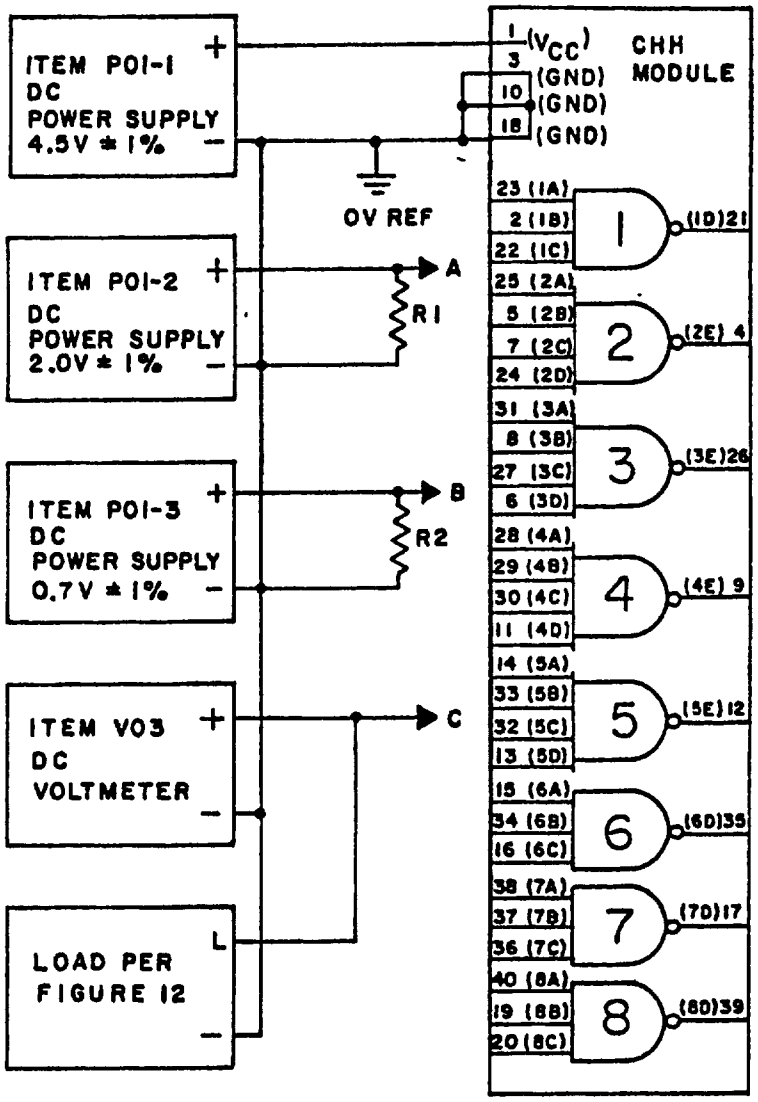
TABLE X. Static output voltage (high-level) test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| H19 | 5E Static output voltage, high level | 13,33,32 | 14 | 12 | 2.5 | 4.5 | V | 11 |
| H20 | 6D | 15,16 | 34 | 35 | | | | |
| H21 | 6D | 15,34 | 16 | 35 | | | | |
| H22 | 6D | 16,34 | 15 | 35 | | | | |
| H23 | 7D | 37,38 | 36 | 17 | | | | |
| H24 | 7D | 36,38 | 37 | 17 | | | | |
| H25 | 7D | 36,37 | 38 | 17 | | | | |
| H26 | 8D | 20,40 | 19 | 39 | | | | |
| H27 | 8D | 19,40 | 20 | 39 | | | | |
| H28 | 8D Static output voltage, high level | 19,20 | 40 | 39 | 2.5 | 4.5 | V | 11 |





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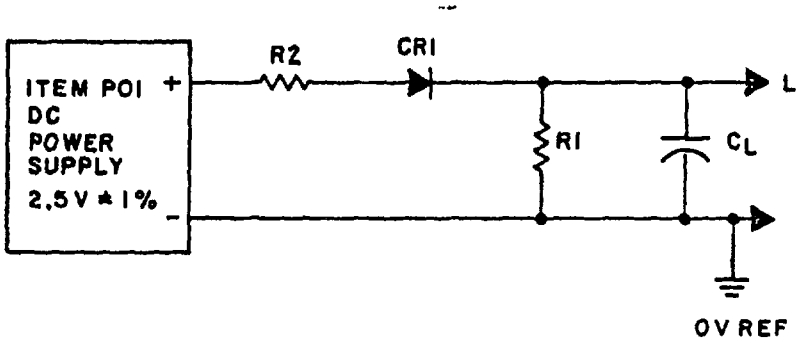



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 11. Static output voltage (low -level and high-level) test circuit for tables IX and X.



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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R2 = Select value for a current flow of 4.0 mA ±5% into point L with point L held at 0.4 V.
3. R1 = 6.19 kΩ ±1%, 1/8 W.
4. Test type S:
   $C_L$ = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   $C_L$ = 150 ±5 pF (including probe and parasitic capacitance) except for crosstalk test which will use 50 ±5 pF.
5. Capacitor $C_L$ is not required for dc electrical test conditions G1-G8 and H1-H28.
6. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 12. CHH logic load circuit.



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TABLE XI. Input current (high-level) test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J1 | 1A Input current, high level | 23 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0 | 20 | µA | 13 |
| J2 | 1B | 2 | 5,6,7,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J3 | 1C | 22 | 2,5,6,7,8,11,13,14,15,16, 19,20,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J4 | 2A | 25 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J5 | 2B | 5 | 2,6,7,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J6 | 2C | 7 | 2,5,6,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J7 | 2D | 24 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J8 | 3A | 31 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,32,33,34,36,37,38,40 | | | | |
| J9 | 3B Input current, high level | 8 | 2,5,6,7,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0 | 20 | µA | 13 |





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TABLE XI. Input current (high-level) test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J10 | 3C Input current, high level | 27 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,28,29,30, 31,32,33,34,36,37,38,40 | 0 | 20 | µA | 13 |
| J11 | 3D | 6 | 2,5,7,8,11,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J12 | 4A | 28 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J13 | 4B | 29 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,30, 31,32,33,34,36,37,38,40 | | | | |
| J14 | 4C | 30 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 31,32,33,34,36,37,38,40 | | | | |
| J15 | 4D | 11 | 2,5,6,7,8,13,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J16 | 5A | 14 | 2,5,6,7,8,11,13,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J17 | 5B | 33 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,34,36,37,38,40 | | | | |
| J18 | 5C Input current, high level | 32 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,33,34,36,37,38,40 | 0 | 20 | µA | 13 |





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TABLE XI. Input current (high-level) test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J19 | 5D Input current, high level | 13 | 2,5,6,7,8,11,14,15,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0 | 20 | µA | 13 |
| J20 | 6A | 15 | 2,5,6,7,8,11,13,14,16,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J21 | 6B | 34 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,36,37,38,40 | | | | |
| J22 | 6C | 16 | 2,5,6,7,8,11,13,14,15,19, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | | | | |
| J23 | 7A | 38 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,40 | | | | |
| J24 | 7B | 37 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,38,40 | | | | |
| J25 | 7C | 36 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,37,38,40 | | | | |
| J26 | 8A | 40 | 2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38 | | | | |
| J27 | 8B Input current, high level | 19 | 2,5,6,7,8,11,13,14,15,16, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0 | 20 | µA | 13 |





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TABLE XI. Input current (high-level) test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J28 | 8C Input current, high level | 20 | 2,5,6,7,8,11,13,14,15,16, 19,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40 | 0 | 20 | µA | 13 |





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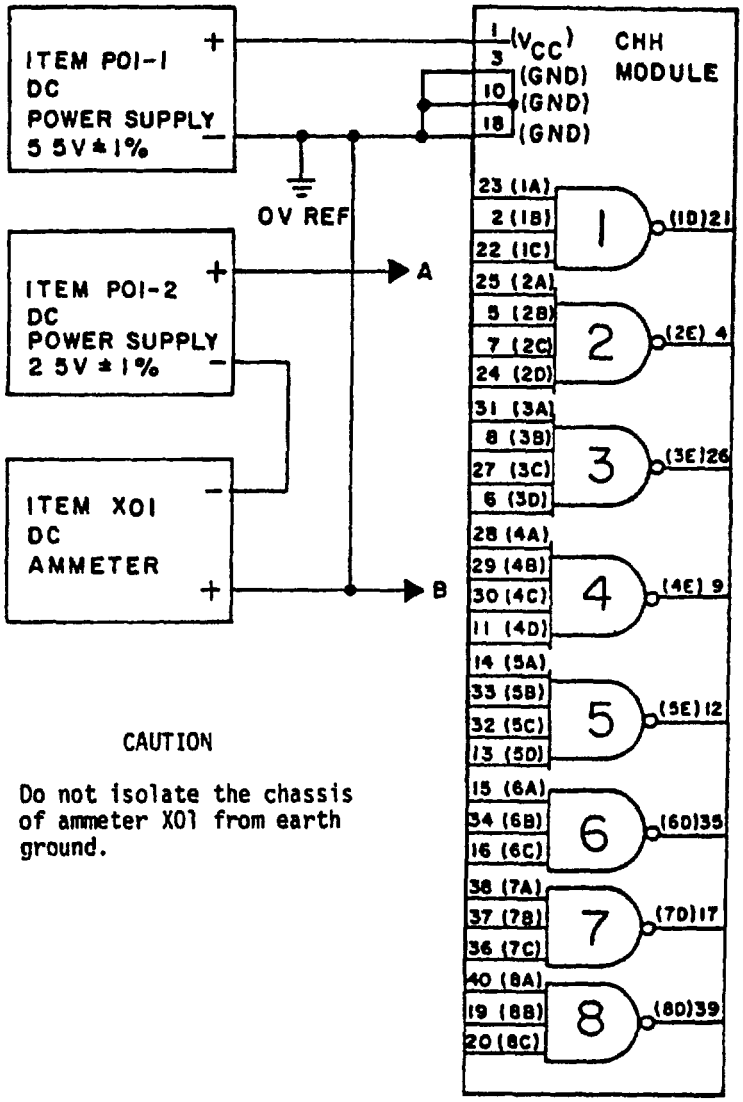



FIGURE 13. Input current (high-level) test circuit for table XI.



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TABLE XII. Input current (low-level) test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| K1 | 1A Input current, low level | 23 | -0.01 | -0.434 | mA | 14 |
| K2 | 1B | 2 | | | | |
| K3 | 1C | 22 | | | | |
| K4 | 2A | 25 | | | | |
| K5 | 2B | 5 | | | | |
| K6 | 2C | 7 | | | | |
| K7 | 2D | 24 | | | | |
| K8 | 3A | 31 | | | | |
| K9 | 3B | 8 | | | | |
| K10 | 3C | 27 | | | | |
| K11 | 3D | 6 | | | | |
| K12 | 4A | 28 | | | | |
| K13 | 4B | 29 | | | | |
| K14 | 4C | 30 | | | | |
| K15 | 4D | 11 | | | | |
| K16 | 5A | 14 | | | | |
| K17 | 5B | 33 | | | | |
| K18 | 5C Input current, low level | 32 | -0.01 | -0.434 | mA | 14 |





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TABLE XII. Input current (low-level) test - Continued.

| Test condition | Parameter | Connect A to pin | Initial limits 25° ±5°C Ambient Min | Initial limits 25° ±5°C Ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|
| K19 | 5D Input current, low level | 13 | -0.01 | -0.434 | mA | 14 |
| K20 | 6A | 15 | | | | |
| K21 | 6B | 34 | | | | |
| K22 | 6C | 16 | | | | |
| K23 | 7A | 38 | | | | |
| K24 | 7B | 37 | | | | |
| K25 | 7C | 36 | | | | |
| K26 | 8A | 40 | | | | |
| K27 | 8B | 19 | | | | |
| K28 | 8C Input current, low level | 20 | -0.01 | -0.434 | mA | 14 |





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Check the source to verify that this is the current version before use.

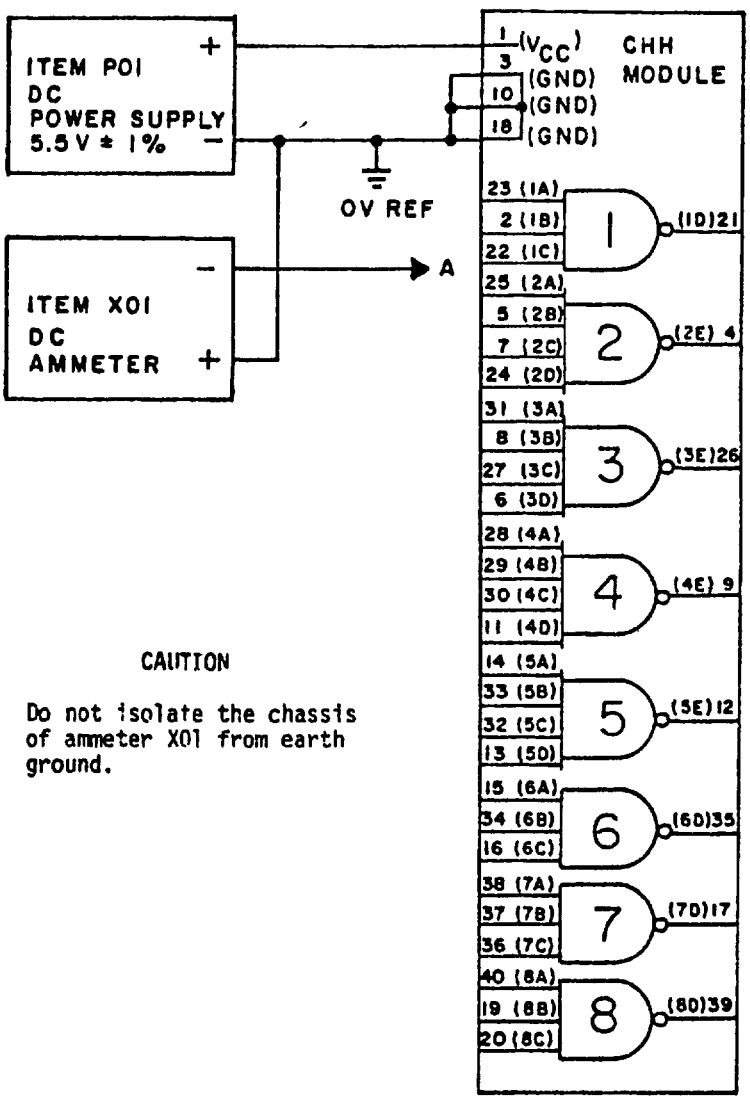



FIGURE 14. Input current (low-level) test circuit for table XII.



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TABLE XIII. Continuity test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| L1 | Continuity | 3 | 0.0 | 1.0 | Ω | 15 |
| L2 | ↓ | 18 | ↓ | 1.0 | ↓ | ↓ |
| L3 | ↓ | 3 | ↓ | 0.1 | ↓ | ↓ |
| L4 | Continuity | 18 | 0.0 | 0.1 | Ω | 15 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:26Z
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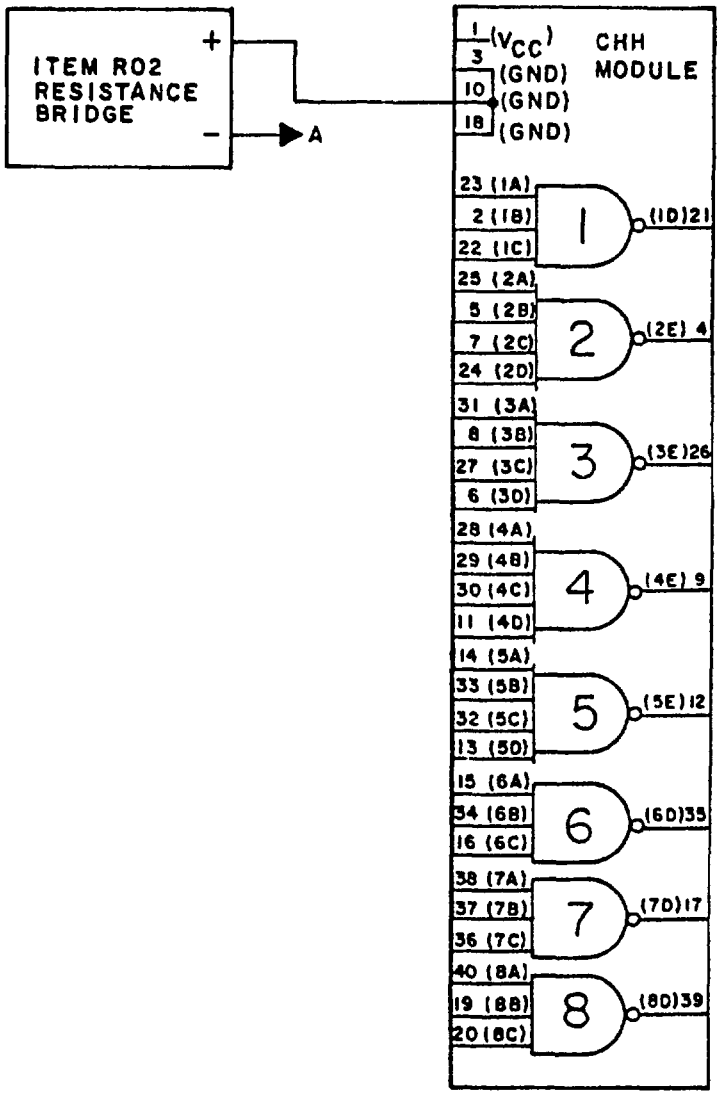



FIGURE 15. Continuity test circuit for table XIII.



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TABLE XIV. Power supply current (low-level outputs and high-level outputs) test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| M1 | Power supply current, low-level outputs | C | Open | 0.1 | 8.8 | mA | 16 |
| M2 | Power supply current, high-level outputs | Open | C | 0.1 | 3.2 | mA | 16 |





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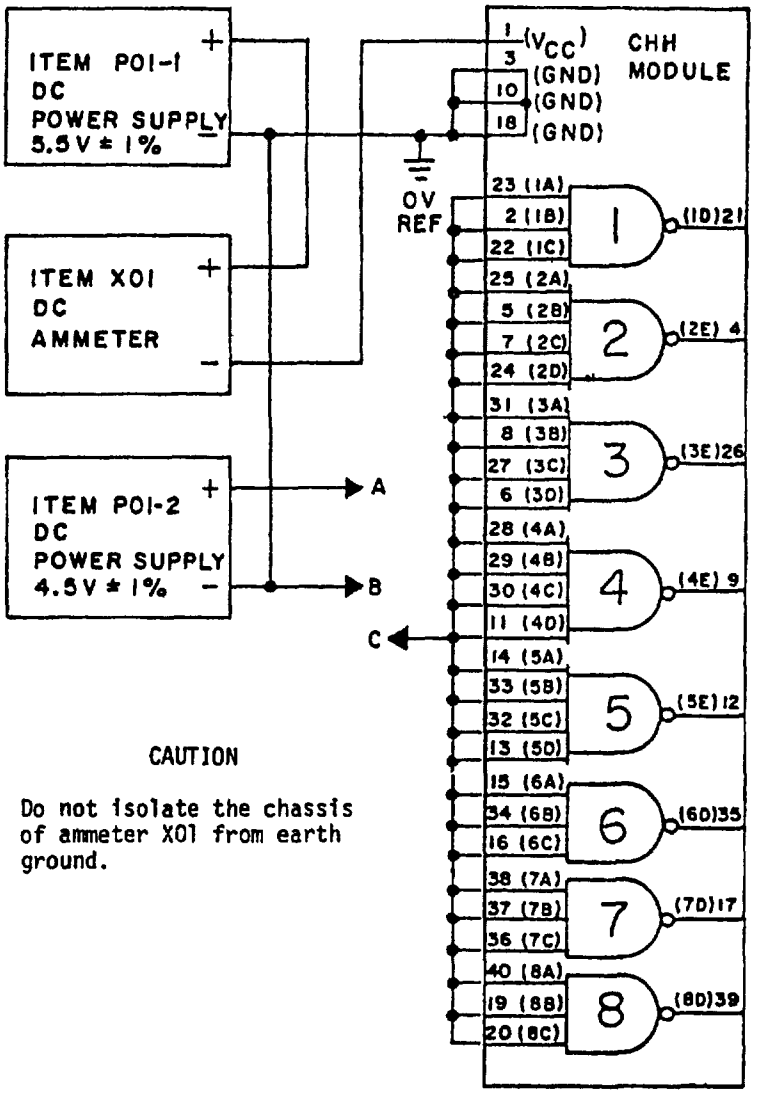



FIGURE 16. Power supply current (low-level outputs and high-level outputs) test circuit for table XIV.



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TABLE XV. Transition time and propagation delay time test.

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| N1 | 1D | Transition time ($t_{TLH}$), $C_L$ = 50 pF | 22,23 | 2 | 21 | 5.0 | 50.0 | ns | 17,18 |
| N2 | 2E | | 24,25,7 | 5 | 4 | | | | |
| N3 | 3E | | 27,31,8 | 6 | 26 | | | | |
| N4 | 4E | | 28,29,30 | 11 | 9 | | | | |
| N5 | 5E | | 32,33,14 | 13 | 12 | | | | |
| N6 | 6D | | 34,16 | 15 | 35 | | | | |
| N7 | 7D | | 37,38 | 36 | 17 | | | | |
| N8 | 8D | Transition time ($t_{TLH}$), $C_L$ = 50 pF | 20,40 | 19 | 39 | 5.0 | 50.0 | | |
| N9 | 1D | Transition time ($t_{TLH}$), $C_L$ = 150 pF | 22,23 | 2 | 21 | 10.0 | 80.0 | | |
| N10 | 2E | | 24,25,7 | 5 | 4 | | | | |
| N11 | 3E | | 27,31,8 | 6 | 26 | | | | |
| N12 | 4E | | 28,29,30 | 11 | 9 | | | | |
| N13 | 5E | | 32,33,14 | 13 | 12 | | | | |
| N14 | 6D | | 34,16 | 15 | 35 | | | | |
| N15 | 7D | | 37,38 | 36 | 17 | | | | |
| N16 | 8D | Transition time ($t_{TLH}$), $C_L$ = 150 pF | 20,40 | 19 | 39 | 10.0 | 80.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| P1 | 1D Transition time ($t_{THL}$), $C_L$ = 50 pF | 22,23 | 2 | 21 | 3.0 | 18.0 | ns | 17,18 |
| P2 | 2E | 24,25,7 | 5 | 4 | | | | |
| P3 | 3E | 27,31,8 | 6 | 26 | | | | |
| P4 | 4E | 28,29,30 | 11 | 9 | | | | |
| P5 | 5E | 32,33,14 | 13 | 12 | | | | |
| P6 | 6D | 34,16 | 15 | 35 | | | | |
| P7 | 7D | 37,38 | 36 | 17 | | | | |
| P8 | 8D Transition time ($t_{THL}$), $C_L$ = 50 pF | 20,40 | 19 | 39 | 3.0 | 18.0 | | |
| P9 | 1D Transition time ($t_{THL}$), $C_L$ = 150 pF | 22,23 | 2 | 21 | 6.0 | 40.0 | | |
| P10 | 2E | 24,25,7 | 5 | 4 | | | | |
| P11 | 3E | 27,31,8 | 6 | 26 | | | | |
| P12 | 4E | 28,29,30 | 11 | 9 | | | | |
| P13 | 5E | 32,33,14 | 13 | 12 | | | | |
| P14 | 6D | 34,16 | 15 | 35 | | | | |
| P15 | 7D | 37,38 | 36 | 17 | | | | |
| P16 | 8D Transition time ($t_{THL}$), $C_L$ = 150 pF | 20,40 | 19 | 39 | 6.0 | 40.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| Q1 | 1A to 1D Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 2,22 | 23 | 21 | 3.0 | 19.0 | ns | 17,18 |
| Q2 | 1B to 1D | 22,23 | 2 | 21 | | | | |
| Q3 | 1C to 1D | 2,23 | 22 | 21 | | | | |
| Q4 | 2A to 2E | 5,7,24 | 25 | 4 | | | | |
| Q5 | 2B to 2E | 7,24,25 | 5 | 4 | | | | |
| Q6 | 2C to 2E | 5,24,25 | 7 | 4 | | | | |
| Q7 | 2D to 2E | 5,7,25 | 24 | 4 | | | | |
| Q8 | 3A to 3E | 8,27,6 | 31 | 26 | | | | |
| Q9 | 3B to 3E | 6,27,31 | 8 | 26 | | | | |
| Q10 | 3C to 3E | 6,8,31 | 27 | 26 | | | | |
| Q11 | 3D to 3E | 8,27,31 | 6 | 26 | | | | |
| Q12 | 4A to 4E | 29,30,11 | 28 | 9 | | | | |
| Q13 | 4B to 4E | 28,30,11 | 29 | 9 | | | | |
| Q14 | 4C to 4E | 28,29,11 | 30 | 9 | | | | |
| Q15 | 4D to 4E | 28,29,30 | 11 | 9 | | | | |
| Q16 | 5A to 5E | 13,33,32 | 14 | 12 | | | | |
| Q17 | 5B to 5E Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 13,14,32 | 33 | 12 | 3.0 | 19.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| Q18 | 5C to 5E Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 13,14,33 | 32 | 12 | 3.0 | 19.0 | ns | 17,18 |
| Q19 | 5D to 5E | 14,33,32 | 13 | 12 | | | | |
| Q20 | 6A to 6D | 16,34 | 15 | 35 | | | | |
| Q21 | 6B to 6D | 16,15 | 34 | 35 | | | | |
| Q22 | 6C to 6D | 15,34 | 16 | 35 | | | | |
| Q23 | 7A to 7D | 37,36 | 38 | 17 | | | | |
| Q24 | 7B to 7D | 38,36 | 37 | 17 | | | | |
| Q25 | 7C to 7D | 37,38 | 36 | 17 | | | | |
| Q26 | 8A to 8D | 19,20 | 40 | 39 | | | | |
| Q27 | 8B to 8D | 20,40 | 19 | 39 | | | | |
| Q28 | 8C to 8D Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 19,40 | 20 | 39 | 3.0 | 19.0 | | |
| Q29 | 1A to 1D Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 2,22 | 23 | 21 | 4.0 | 29.0 | | |
| Q30 | 1B to 1D | 22,23 | 2 | 21 | | | | |
| Q31 | 1C to 1D | 2,23 | 22 | 21 | | | | |
| Q32 | 2A to 2E | 5,7,24 | 25 | 4 | | | | |
| Q33 | 2B to 2E Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 7,24,25 | 5 | 4 | 4.0 | 29.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| Q34 | 2C to 2E Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 5,24,25 | 7 | 4 | 4.0 | 29.0 | ns | 17,18 |
| Q35 | 2D to 2E | 5,7,25 | 24 | 4 | | | | |
| Q36 | 3A to 3E | 8,27,6 | 31 | 26 | | | | |
| Q37 | 3B to 3E | 31,27,6 | 8 | 26 | | | | |
| Q38 | 3C to 3E | 31,8,6 | 27 | 26 | | | | |
| Q39 | 3D to 3E | 31,8,27 | 6 | 26 | | | | |
| Q40 | 4A to 4E | 29,30,11 | 28 | 9 | | | | |
| Q41 | 4B to 4E | 28,30,11 | 29 | 9 | | | | |
| Q42 | 4C to 4E | 28,29,11 | 30 | 9 | | | | |
| Q43 | 4D to 4E | 28,29,30 | 11 | 9 | | | | |
| Q44 | 5A to 5E | 13,33,32 | 14 | 12 | | | | |
| Q45 | 5B to 5E | 13,14,32 | 33 | 12 | | | | |
| Q46 | 5C to 5E | 13,14,33 | 32 | 12 | | | | |
| Q47 | 5D to 5E | 14,33,32 | 13 | 12 | | | | |
| Q48 | 6A to 6D | 34,16 | 15 | 35 | | | | |
| Q49 | 6B to 6D | 15,16 | 34 | 35 | | | | |
| Q50 | 6C to 6D Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 15,34 | 16 | 35 | 4.0 | 29.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25°C ± 5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| Q51 | 7A to 7D Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 36,37 | 38 | 17 | 4.0 | 29.0 | ns | 17,18 |
| Q52 | 7B to 7D | 36,38 | 37 | 17 | | | | |
| Q53 | 7C to 7D | 37,38 | 36 | 17 | | | | |
| Q54 | 8A to 8D | 19,20 | 40 | 39 | | | | |
| Q55 | 8B to 8D | 20,40 | 19 | 39 | | | | |
| Q56 | 8C to 8D Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 19,40 | 20 | 39 | 4.0 | 29.0 | | |
| R1 | 1A to 1D Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 2,22 | 23 | 21 | 3.0 | 19.0 | | |
| R2 | 1B to 1D | 22,23 | 2 | 21 | | | | |
| R3 | 1C to 1D | 2,23 | 22 | 21 | | | | |
| R4 | 2A to 2E | 5,7,24 | 25 | 4 | | | | |
| R5 | 2B to 2E | 7,24,25 | 5 | 4 | | | | |
| R6 | 2C to 2E | 5,24,25 | 7 | 4 | | | | |
| R7 | 2D to 2E | 5,7,25 | 24 | 4 | | | | |
| R8 | 3A to 3E | 8,27,6 | 31 | 26 | | | | |
| R9 | 3B to 3E | 6,27,31 | 8 | 26 | | | | |
| R10 | 3C to 3E Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 6,8,31 | 27 | 26 | 3.0 | 19.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| R11 | 3D to 3E Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 8,27,31 | 6 | 26 | 3.0 | 19.0 | ns | 17,18 |
| R12 | 4A to 4E | 29,30,11 | 28 | 9 | | | | |
| R13 | 4B to 4E | 28,30,11 | 29 | 9 | | | | |
| R14 | 4C to 4E | 28,29,11 | 30 | 9 | | | | |
| R15 | 4D to 4E | 28,29,30 | 11 | 9 | | | | |
| R16 | 5A to 5E | 13,33,32 | 14 | 12 | | | | |
| R17 | 5B to 5E | 13,14,32 | 33 | 12 | | | | |
| R18 | 5C to 5E | 13,14,33 | 32 | 12 | | | | |
| R19 | 5D to 5E | 14,33,32 | 13 | 12 | | | | |
| R20 | 6A to 6D | 16,34 | 15 | 35 | | | | |
| R21 | 6B to 6D | 15,16 | 34 | 35 | | | | |
| R22 | 6C to 6D | 15,34 | 16 | 35 | | | | |
| R23 | 7A to 7D | 37,36 | 38 | 17 | | | | |
| R24 | 7B to 7D | 38,36 | 37 | 17 | | | | |
| R25 | 7C to 7D | 37,38 | 36 | 17 | | | | |
| R26 | 8A to 8D | 19,20 | 40 | 39 | | | | |
| R27 | 8B to 8D | 20,40 | 19 | 39 | | | | |
| R28 | 8C to 8D Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 19,40 | 20 | 39 | 3.0 | 19.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| R29 | 1A to 1D Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 2,22 | 23 | 21 | 4.0 | 29.0 | ns | 17,18 |
| R30 | 1B to 1D | 22,23 | 2 | 21 | | | | |
| R31 | 1C to 1D | 2,23 | 22 | 21 | | | | |
| R32 | 2A to 2E | 5,7,24 | 25 | 4 | | | | |
| R33 | 2B to 2E | 7,24,25 | 5 | 4 | | | | |
| R34 | 2C to 2E | 5,24,25 | 7 | 4 | | | | |
| R35 | 2D to 2E | 5,7,25 | 24 | 4 | | | | |
| R36 | 3A to 3E | 8,27,6 | 31 | 26 | | | | |
| R37 | 3B to 3E | 31,27,6 | 8 | 26 | | | | |
| R38 | 3C to 3E | 31,8,6 | 27 | 26 | | | | |
| R39 | 3D to 3E | 31,8,27 | 6 | 26 | | | | |
| R40 | 4A to 4E | 29,30,11 | 28 | 9 | | | | |
| R41 | 4B to 4E | 28,30,11 | 29 | 9 | | | | |
| R42 | 4C to 4E | 28,29,11 | 30 | 9 | | | | |
| R43 | 4D to 4E | 28,29,30 | 11 | 9 | | | | |
| R44 | 5A to 5E | 13,33,32 | 14 | 12 | | | | |
| R45 | 5B to 5E Propagetion delay time ($t_{PHL}$), $C_L$ = 150 pF | 13,14,32 | 33 | 12 | 4.0 | 29.0 | ns | 17,18 |





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TABLE XV. Transition time and propagation delay time test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| R46 | 5C to 5E Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 13,14,33 | 32 | 12 | 4.0 | 29.0 | ns | 17,18 |
| R47 | 5D to 5E | 14,33,32 | 13 | 12 | | | | |
| R48 | 6A to 6D | 34,16 | 15 | 35 | | | | |
| R49 | 6B to 6D | 15,16 | 34 | 35 | | | | |
| R50 | 6C to 6D | 15,34 | 16 | 35 | | | | |
| R51 | 7A to 7D | 36,37 | 38 | 17 | | | | |
| R52 | 7B to 7D | 36,38 | 37 | 17 | | | | |
| R53 | 7C to 7D | 37,38 | 36 | 17 | | | | |
| R54 | 8A to 8D | 19,20 | 40 | 39 | | | | |
| R55 | 8B to 8D | 20,40 | 19 | 39 | | | | |
| R56 | 8C to 8D Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 19,40 | 20 | 39 | 4.0 | 29.0 | ns | 17,18 |





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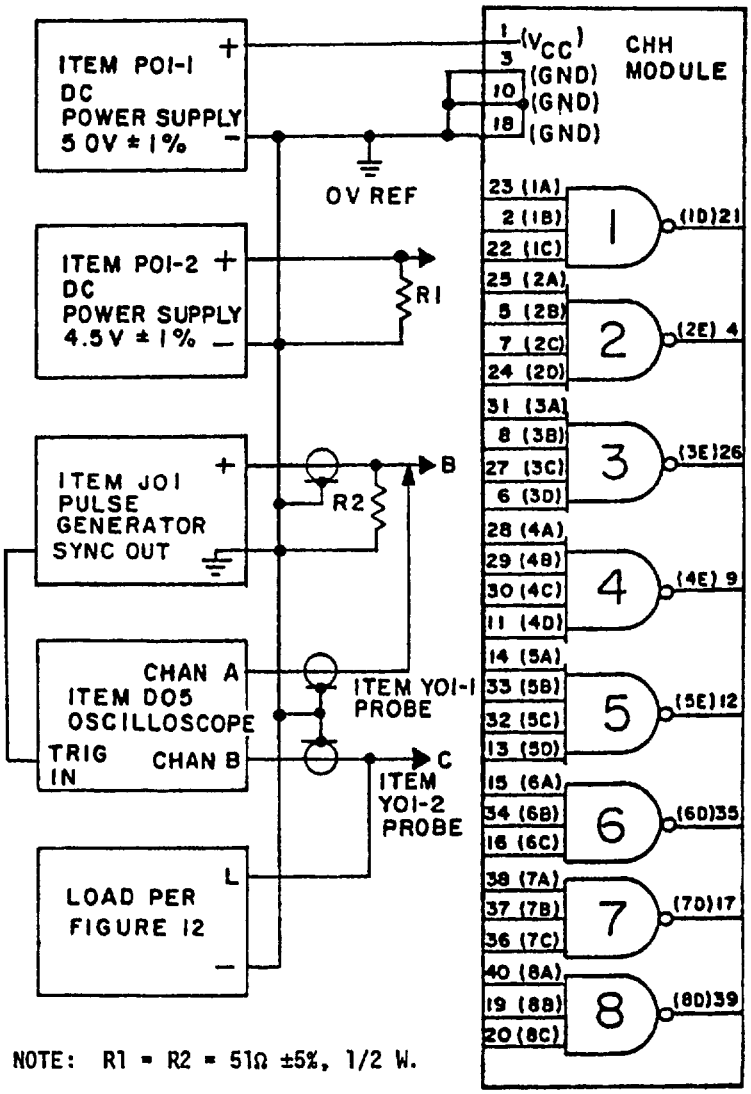
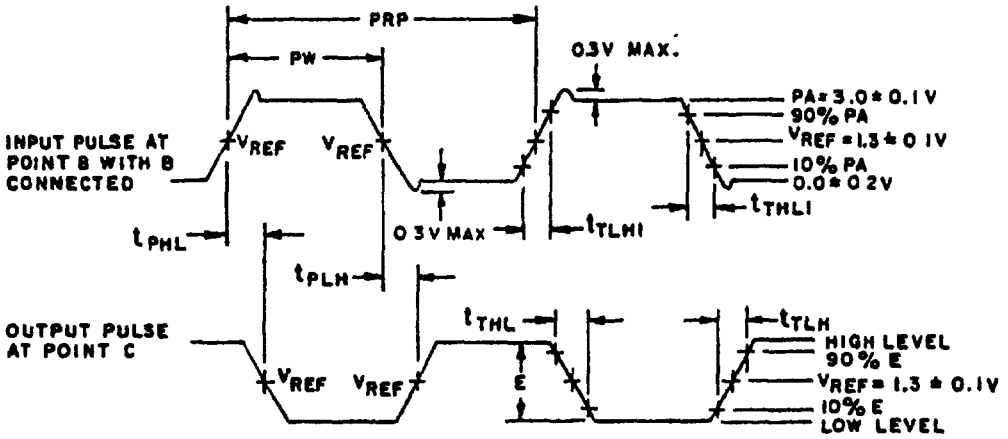



FIGURE 17. Transition time and propagation delay time test circuit for table XV.



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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
   PW = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R2 and and input pins; between probes YO1-1 and YO1-2 and input and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 18. Transition time and propagation delay time test circuit waveforms for table XV.



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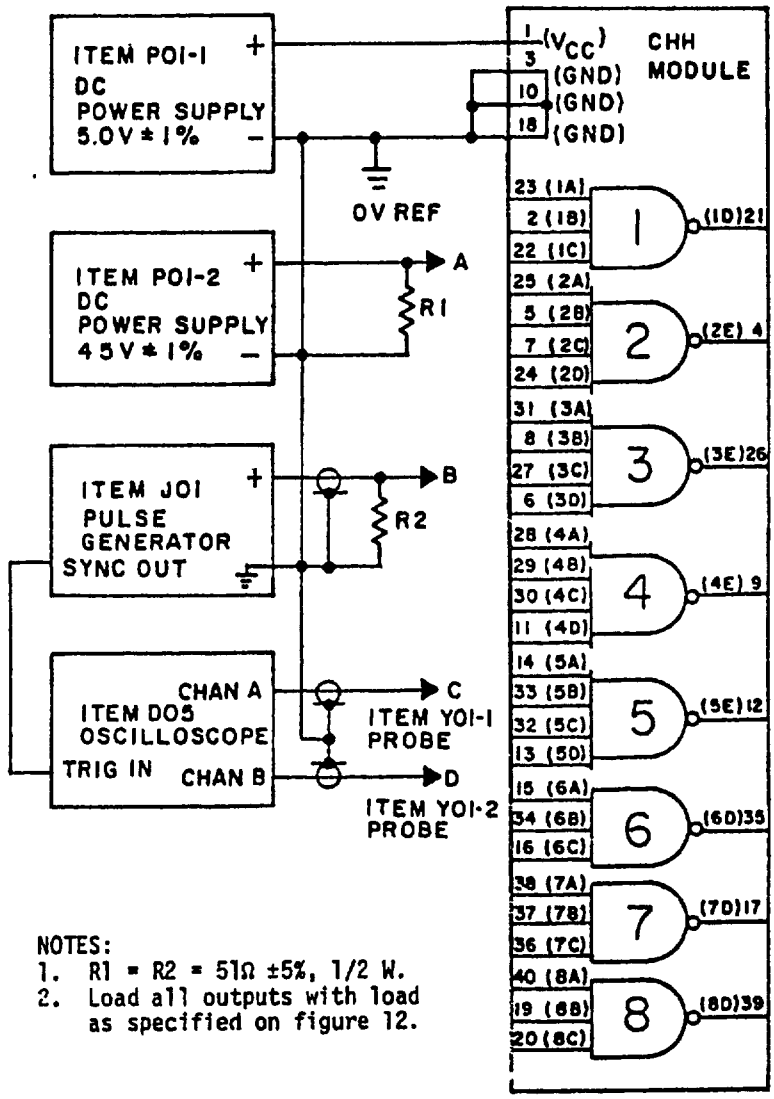
TABLE XVI. Output to output delta propagation delay time ( $t_{PLH}$ and $t_{PHL}$) test.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| S1 | 1D to 2E Output to output delta propagation delay time ( $t_{PLH}$), $C_L$ = 50 pF | 23,2, 24,5,7 | 22,25 | 21 | 4 | -3.0 | 3.0 | ns | 19,20 |
| S2 | 3E to 4E | 6,8,27, 28,29, 11 | 30,31 | 9 | 26 | | | | |
| S3 | 5E to 6D | 32,33, 13,34, 16 | 14,15 | 12 | 35 | | | | |
| S4 | 7D to 8D Output to output delta propagation delay time ( $t_{PLH}$), $C_L$ = 50 pF | 37,36, 19,20 | 38,40 | 17 | 39 | | | | |
| T1 | 1D to 2E Output to output delta propagation delay time ( $t_{PHL}$), $C_L$ = 50 pF | 23,2, 24,5,7 | 22,25 | 21 | 4 | | | | |
| T2 | 3E to 4E | 6,8,27, 28,29, 11 | 30,31 | 9 | 26 | | | | |
| T3 | 5E to 6D | 32,33, 34,13, 16 | 14,15 | 12 | 35 | | | | |
| T4 | 7D to 8D Output to output delta propagation delay time ( $t_{PHL}$), $C_L$ = 50 pF | 37,36, 19,20 | 38,40 | 17 | 39 | -3.0 | 3.0 | ns | 19,20 |





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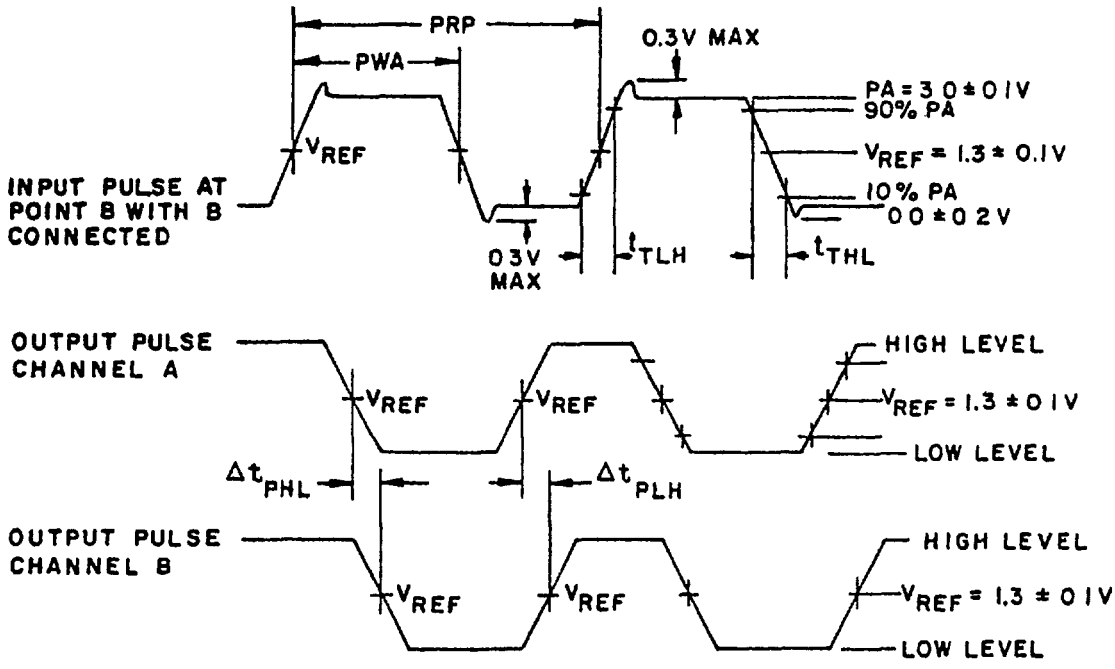



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 12.

FIGURE 19. Output to output delta propagation delay time ($\Delta t_{PLH}$ and $\Delta t_{PHL}$) test circuit for table XVI.



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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. Before making the measurements specified in table XVI, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $<$ 0.4 V.
3. The lead length of electrical conductors between resistor R2 and input pins; between probes YO1-1 and YO1-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 20. Output to output delta propagation delay time ($\Delta t_{PLH}$ and $\Delta t_{PHL}$) test circuit waveforms for table XVI.



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TABLE XVII. Output crosstalk (low-level and high-level) test.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| U1 | 1D Output crosstalk, low level | 2,22, 23,5,7, 24,8, 27,6, 29,30, 11,33, 32,13, 34,16, 37,36, 19,20 | 25,31, 28,14, 15,38, 40 | 21 | Open | 0 | 500 | mV | 21,22 |
| U2 | 2E Output crosstalk, low level | 5,7,24, 25,2, 22,8, 27,6, 29,30, 11,33, 32,13, 34,16, 37,36, 19,20 | 23,31, 28,14, 15,38, 40 | 4 | Open | 0 | 500 | mV | 21,22 |
| U3 | 3E Output crosstalk, low level | 6,8,27 31,2, 22,5,7, 24,29, 30,11, 33,32, 13,34, 16,37, 36,19, 20 | 23,25, 28,14, 15,38, 40 | 26 | Open | 0 | 500 | mV | 21,22 |





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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| U4 | 4E Output crosstalk, low level | 28,29, 30,11, 2,22,5, 7,24,8, 27,6, 33,32, 13,34, 16,37, 36,19, 20 | 23,25, 31,14, 15,38, 40 | 9 | Open | 0 | 500 | mV | 21,22 |
| U5 | 5E Output crosstalk, low level | 13,14, 32,33, 2,22,5, 7,24,8, 27,6, 29,30, 11,34, 16,37, 36,19, 20 | 23,25, 31,28, 15,38, 40 | 12 | Open | 0 | 500 | mV | 21,22 |
| U6 | 6D Output crosstalk, low level | 16,15, 34,2, 22,5,7, 24,8, 27,6, 29,30, 11,33, 32,13, 37,36, 19,20 | 23,25, 31,28, 14,38, 40 | 35 | Open | 0 | 500 | mV | 21,22 |





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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| U7 | 7D Output crosstalk, low level | 36,37, 38,2, 22,5,7, 24,8, 27,6, 29,30, 11,33, 32,13, 34,16, 19,20 | 23,25, 31,28, 14,15, 40 | 17 | Open | 0 | 500 | mV | 21,22 |
| U8 | 8D Output crosstalk, low level | 20,40, 19,2, 22,5,7, 24,8, 27,6, 29,30, 11,33, 32,13, 34,37, 36,16 | 23,25, 31,28, 14,15, 38 | 39 | Open | 0 | 500 | mV | 21,22 |
| V1 | 1D Output crosstalk, high level | 5,7,24, 8,27,6, 29,30, 11,33, 32,13, 34,16, 37,36, 19,20 | 25,31, 28,14, 15,38, 40 | 21 | 2,22, 23 | 0 | 400 | mV | 21,22 |





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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| V2 | 2E Output crosstalk, high level | 2,22,8, 27,6, 29,30, 11,33, 32,13, 34,16, 37,36, 19,20 | 23,31, 28,14, 15,38, 40 | 4 | 24,25, 5,7 | 0 | 400 | mV | 21,22 |
| V3 | 3E | 2,22,5, 7,24, 29,30, 11,33, 32,13, 34,16, 37,36, 19,20 | 23,25, 28,14, 15,38, 40 | 26 | 6,8, 27,31 | | | | |
| V4 | 4E | 2,22,5, 7,24,8, 27,6, 33,32, 13,34, 16,37, 36,19, 20 | 23,25, 31,14, 15,38, 40 | 9 | 11,28, 29,30 | | | | |
| V5 | 5E Output crosstalk, high level | 2,22,5, 7,24,8, 27,6, 29,30, 11,34, 16,37, 36,19, 20 | 23,25, 31,28, 15,38, 40 | 12 | 13,14, 32,33 | 0 | 400 | mV | 21,22 |





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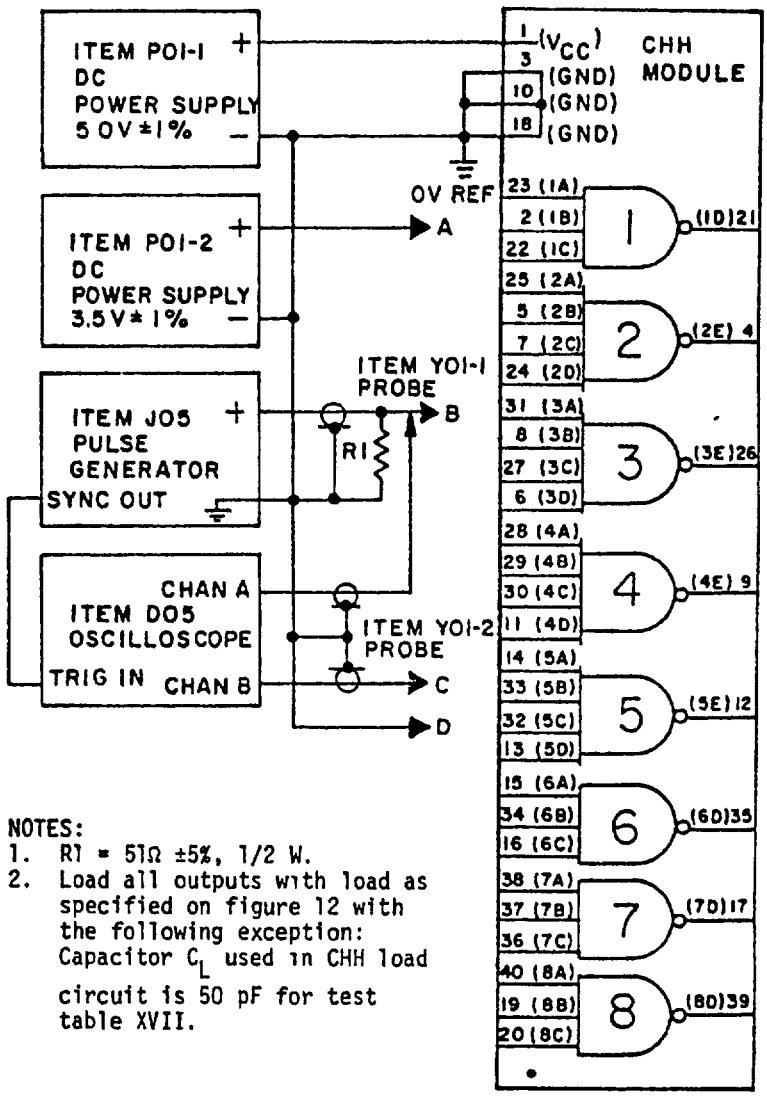
TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| V6 | 6D Output crosstalk, high level | 2,22,5, 7,24,8, 27,6, 29,30, 11,33, 32,13, 37,36, 19,20 | 23,25, 31,28, 14,38, 40 | 35 | 15,16, 34 | 0 | 400 | mV | 21,22 |
| V7 | 7D Output crosstalk, high level | 2,22,5, 7,24,8, 27,6, 29,30, 11,33, 32,13, 34,16, 19,20 | 23,25, 31,28, 14,15, 40 | 17 | 36,37, 38 | 0 | 400 | mV | 21,22 |
| V8 | 8D Output crosstalk, high level | 2,22,5, 7,24,8, 27,6, 29,30, 11,33, 32,13, 34,16, 37,36 | 23,25, 31,28, 14,15, 38 | 39 | 19,20, 40 | 0 | 400 | mV | 21,22 |





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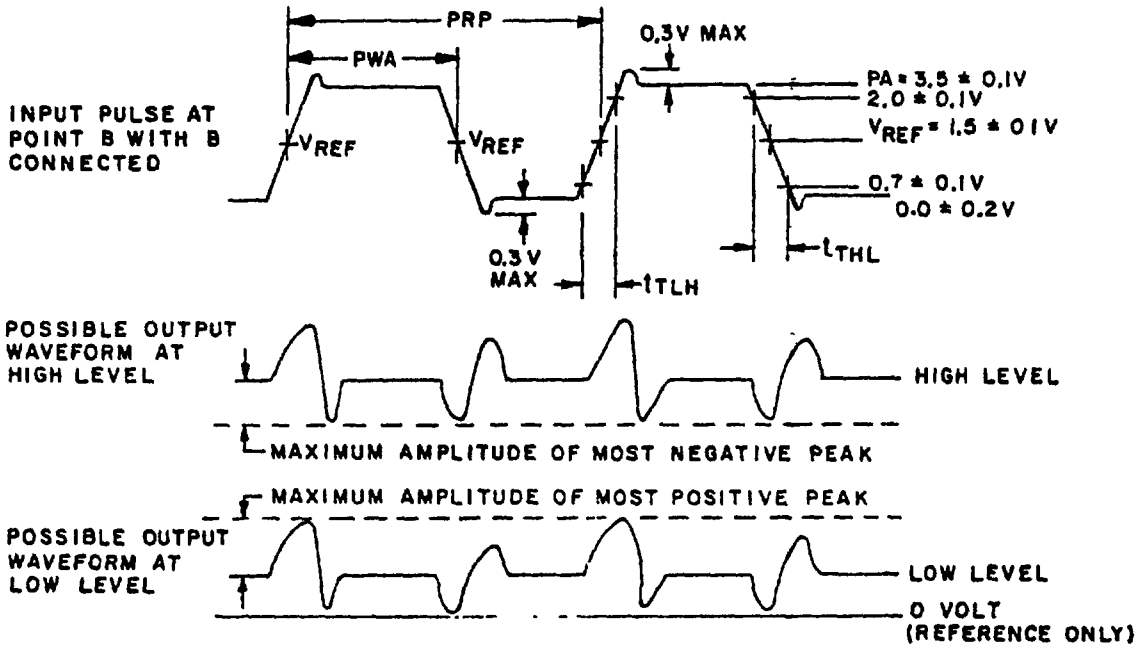



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 12 with the following exception: Capacitor $C_L$ used in CHH load circuit is 50 pF for test table XVII.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit for table XVII.



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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin)
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. The lead length of electrical conductors between resistor RI and input pins, between probes YOI-1 and YOI-2 and input and output load circuits and output pins; and between ground leads of probes YOI-1 and YOI-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

FIGURE 22. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.



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TABLE XVIII. Power supply transient amplitude test.

| Test condition | Parameter | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| W | Power supply transient amplitude | 0 | 400 | mV | 23,24 |





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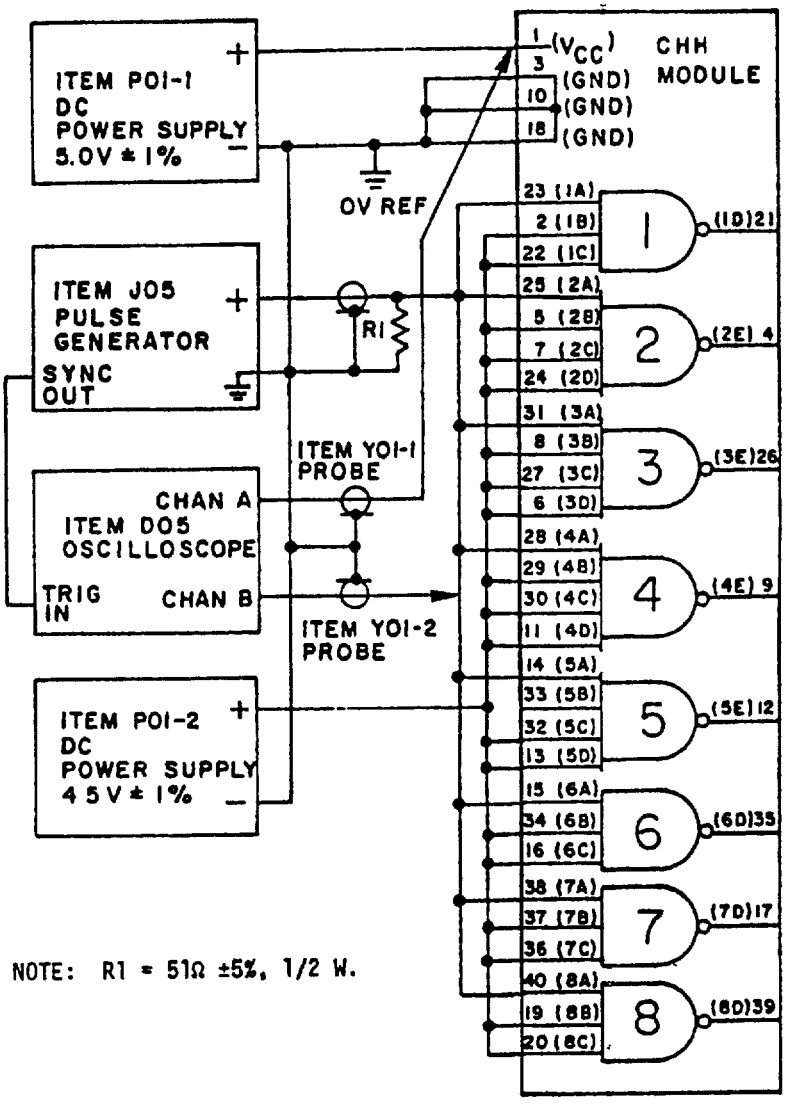
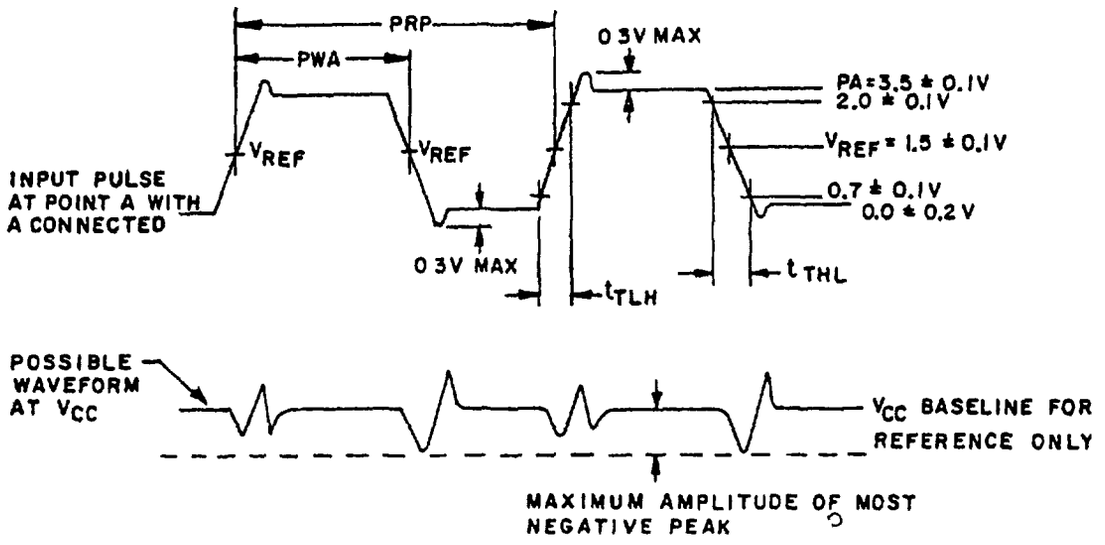



FIGURE 23. Power supply transient amplitude test circuit for table XVIII.



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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns
Verify $t_{TLH}$ from 5.0 to 25.0 ns and $t_{THL}$ from 3.0 to 10.0 ns and record the maximum value of transient amplitude.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.



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TABLE XIX. Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| X | Power supply filter capacitance | Open | 1 | 1.6 | 3.0 | µF | 25 |





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TABLE XX. Input capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| Y1 | 1A Input capacitance | 23 | Open | 5.0 | 15.0 | pF | 25 |
| Y2 | 1B | 2 | | | | | |
| Y3 | 1C | 22 | | | | | |
| Y4 | 2A | 25 | | | | | |
| Y5 | 2B | 5 | | | | | |
| Y6 | 2C | 7 | | | | | |
| Y7 | 2D | 24 | | | | | |
| Y8 | 3A | 31 | | | | | |
| Y9 | 3B | 8 | | | | | |
| Y10 | 3C | 27 | | | | | |
| Y11 | 3D | 6 | | | | | |
| Y12 | 4A | 28 | | | | | |
| Y13 | 4B | 29 | | | | | |
| Y14 | 4C | 30 | | | | | |
| Y15 | 4D | 11 | | | | | |
| Y16 | 5A | 14 | | | | | |
| Y17 | 5B | 33 | | | | | |
| Y18 | 5C Input capacitance | 32 | Open | 5.0 | 15.0 | pF | 25 |





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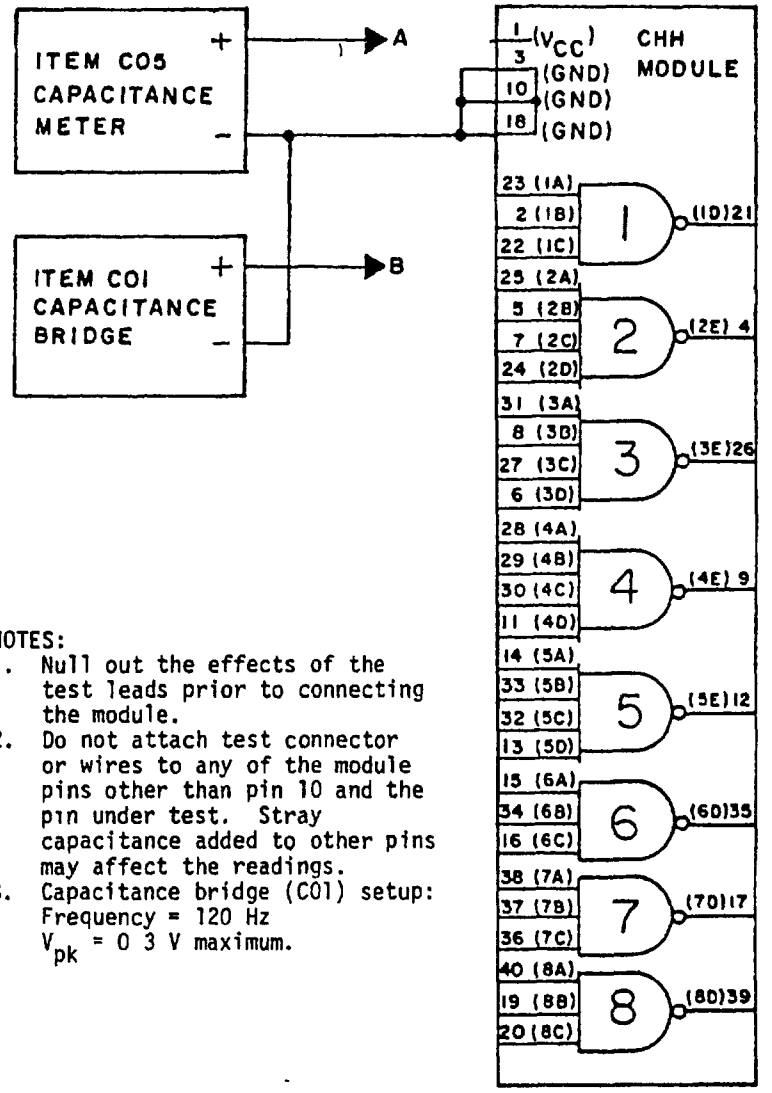
TABLE XX. Input capacitance test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| Y19 | 5D Input capacitance | 13 | Open | 5.0 | 15.0 | pF | 25 |
| Y20 | 6A | 15 | | | | | |
| Y21 | 6B | 34 | | | | | |
| Y22 | 6C | 16 | | | | | |
| Y23 | 7A | 38 | | | | | |
| Y24 | 7B | 37 | | | | | |
| Y25 | 7C | 36 | | | | | |
| Y26 | 8A | 40 | | | | | |
| Y27 | 8B | 19 | | | | | |
| Y28 | 8C Input capacitance | 20 | Open | 5.0 | 15.0 | pF | 25 |





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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup: Frequency = 120 Hz
   $V_{pk}$ = 0 3 V maximum.

FIGURE 25. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.



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